Chapter - 1

Communication = sharing information

local, remote

Data = information / message

↳ Agreed upon by the parties (which data will accept or reject pre decide)

Data / Message

Sender

Reciever

medium

Protocol

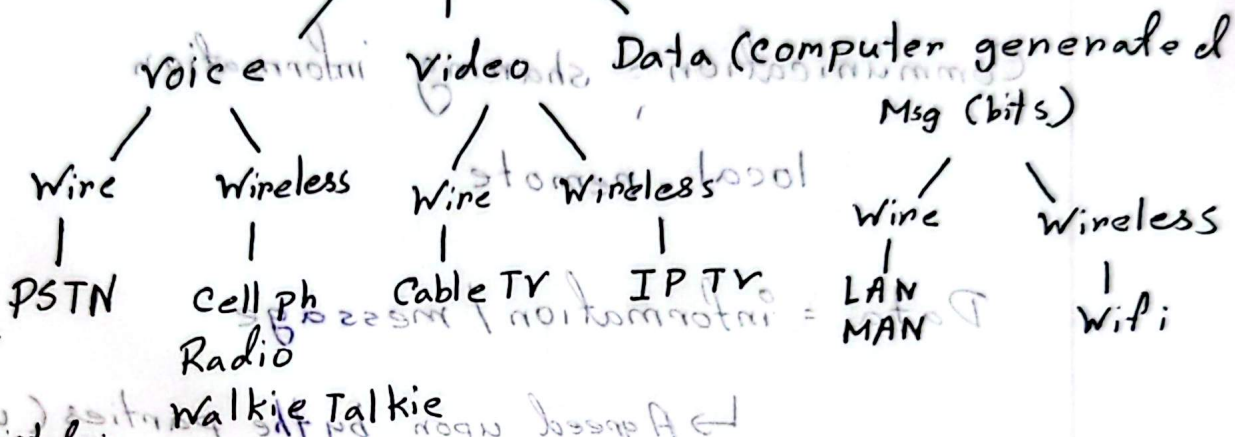
Data (Computer generated msg (bits))

unicode

Different type of data কিভাবে represent করা হয়?

Comp

Data Comistgaid



Medium: Twisted wire

Describe ১, Wired device which (transmits Data) communication describe: components ~~compo device~~

Senden প্রাপকের

reciever

medium: wire

protocal: Describe করবে আগামীকাল,
Data

L-2

i) Defination of Data Com

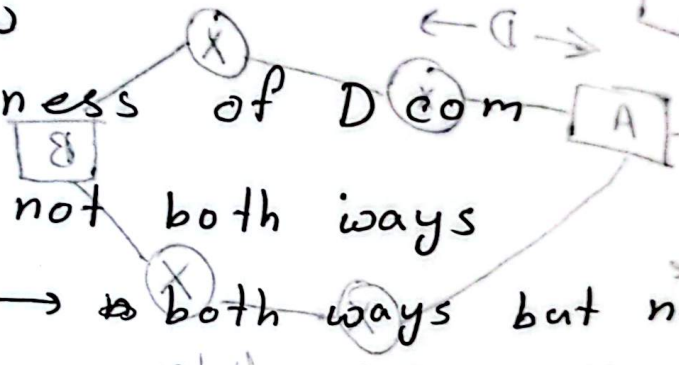
ii) Components of Data Com — Sender, Receiver, Transmission medium, MSG, protocol

iii) Data representation

iv) Categories of Data Com

v) Data flow

vi) Effectiveness of D com



Simplex → not both ways

Half Duplex → both ways but not at a time

Full Duplex → both ways togather

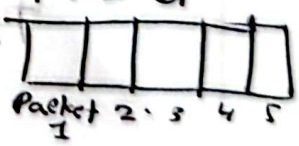
Delivery ensures by address system.

Accuracy : Noise eancle

Timeliness : same time ↺ audio and video

$\frac{D}{2}$ send

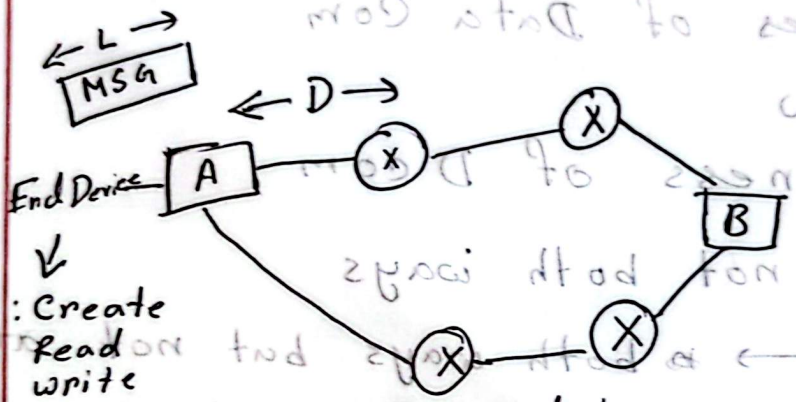
Jitter → uneven time ↺ packet আসার এ delay ↺ jitter



1) Network (for large scale communication)

End Device | Intermediate device

link



MSG : Create
Read
write

যুক্ত end device

OSI model এর 7 layer access করতে পারে

Intermediate device

OSI model এর 3rd layer access করে only forwards

Performance measure হয় delay দিয়ে

Propagation delay = $\frac{D}{S}$ sec = $\frac{D}{3\times10^8 m/s}$

D ← link distance

S ← speed

Transmission delay = $\frac{L}{R}$ sec

L ← Msg size

R Data rate

mbps
kbps

response time : Website এ ঢুকার time এ

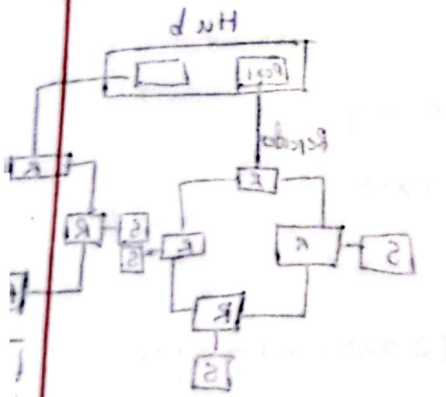
http server এর response time

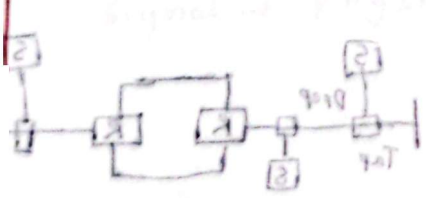
* Reliability:

কত fast recover করা যায়

Security: encryption

Bus → multipoint



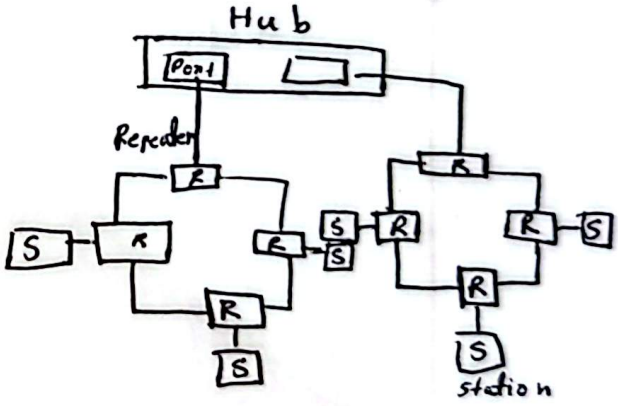


1) Network defination

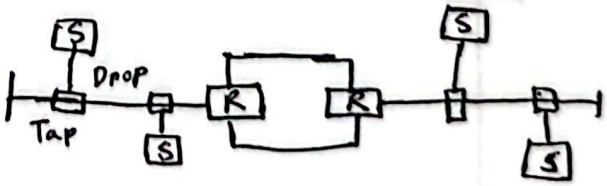
2) Connection type — Point to point
                    — Multipoint

3) Topology — Mesh
            — star
            — Ring
            — Bus

4) Hybrid topology — ① star with bus
                   ② star with ring (2)



③ Ring with bus



802.11 (protocol of how wifi will work)

Passband channel ... analogue conversion

Digital data
Analog data → Analog signal

Bandwidth, BW ≠ 0

Digital Data → Analog signal

→ sine wave (carrier freq, f_c)

→ $s(t) = A \sin(2\pi f_c t + \phi)$

Amp (ASK) / Freq (FSK) / Phase (PSK)

(1) Implementation process
(2) Bandwidth → ... को सीधे ... convert ...

Improved version → QAM

bits এর 0 or 1 থাকলে e এর একটা factor ছিলো। But এখানে নেই

$$r = \frac{\text{number of data elements}}{\text{number of signal elements}}$$

for, ASK, FSK, PSK, number of signal element = 1

$$r = \frac{\text{number of data element}}{1}$$

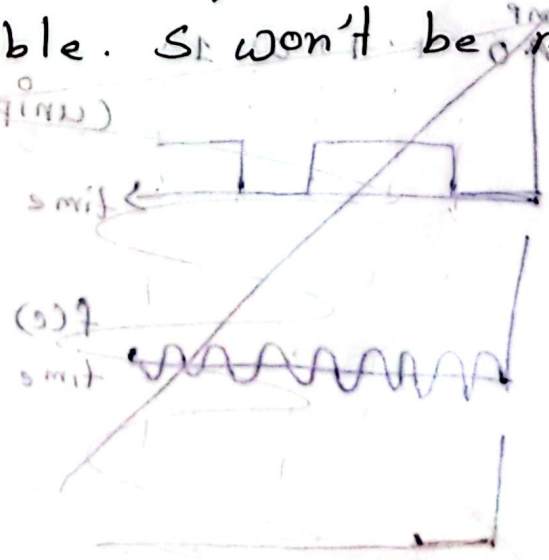
$$S = N \times \frac{1}{r}$$

$r = 1$; $S = N$ (×) ← $r \uparrow$ $S \downarrow$

$r = 2$; $S < N$ ↑

we want less signal element to go. many data through less signal element.

In this conversion ~~tegnig~~ technique this is achievable. S won't be more than 1.



ASK

Date elem : 1
signal " : 1

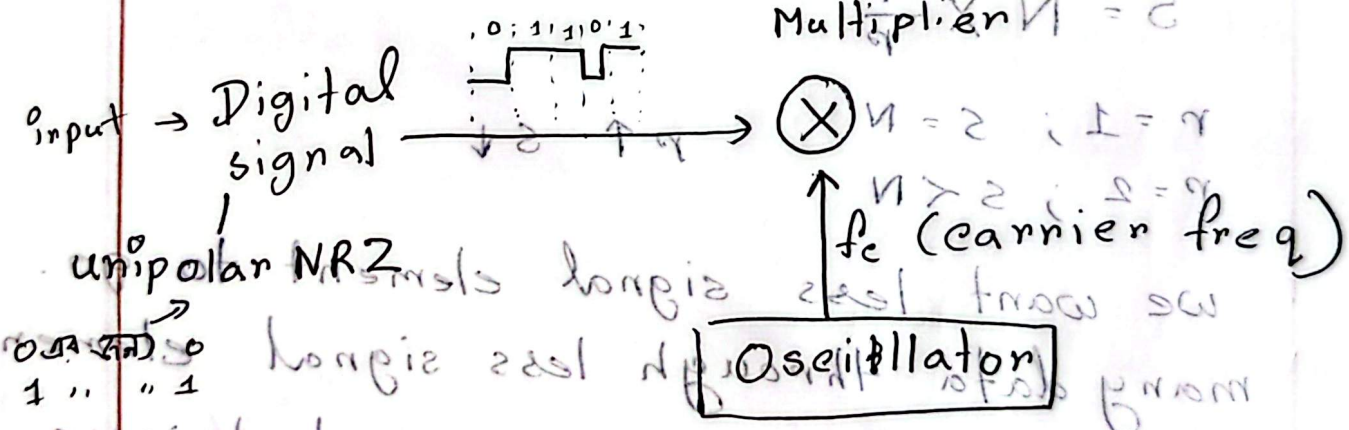
only number of bit 1 works on ASK. so we call it BASK

$n_b = 1$

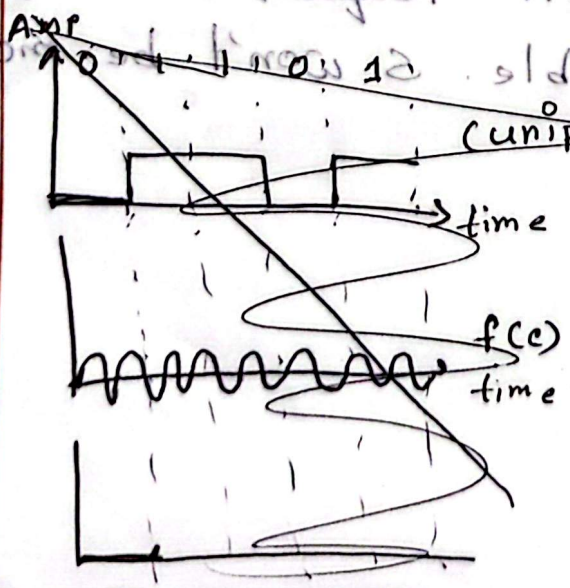
$2^{n_b} = 2^1 = 2$ — 1, 0

BASK (only binary করা যায়)

1 এর জন্য 3টা frequency



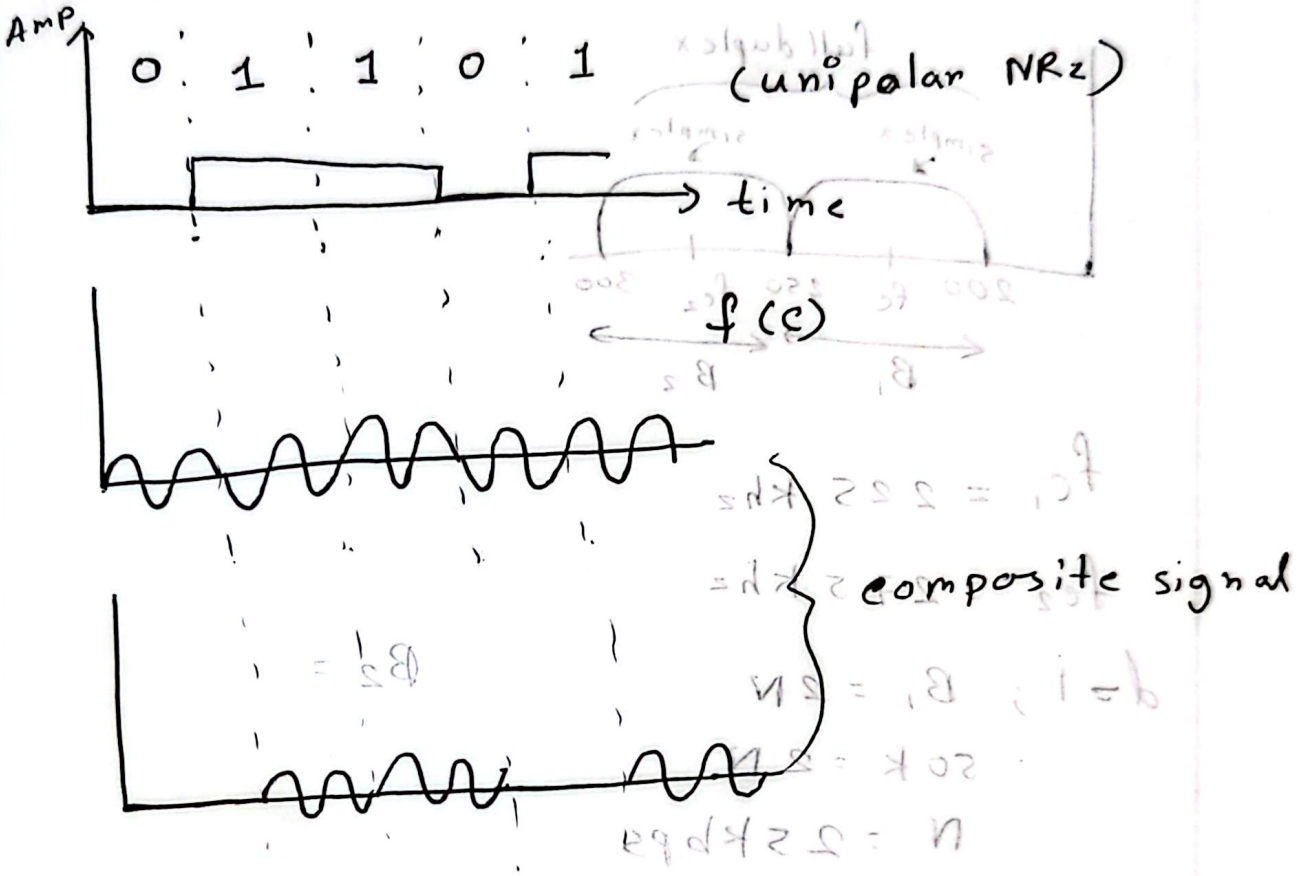
$fc = A \sin(2\pi f t + \phi)$



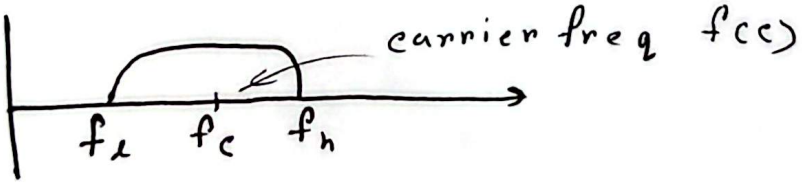
1 bit interval ↓
2 or 3 cycle

Alarm periodic → discrete (f domain)

Voice non " → continuous (f dom)



BW



BASK, r = 1, S = N

$$S = N \times \frac{1}{r} = N$$

$$f_l = f_c - \frac{S}{2} = f_c - \frac{N}{2}$$ ✓

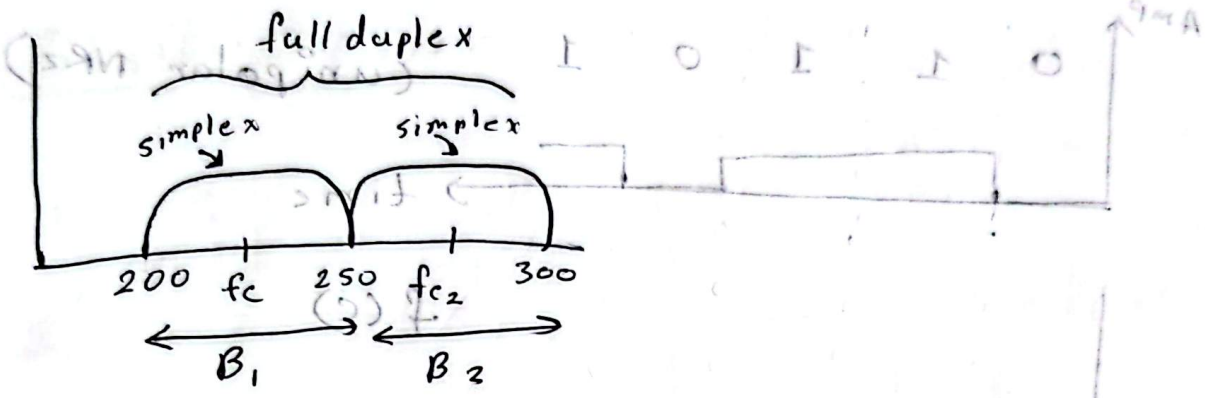
$$f_h = f_c + \frac{S}{2} = f_c + \frac{N}{2}$$ ✓

Bandwidth $\propto$ S

$$B = (1+d)S$$

$d = 0$, $B_{min} = (1+0)S = N$ ✓

$d = 1$, $B_{max} = (1+1)S = 2N$ ✓



$f_{c_1} = 225\ kh_z$

$f_{c_2} = 275\ kh_z$

$d = 1$; $B_1 = 2N$ $\qquad B_2 =$

$50k = 2N$

$N = 25kbps$

Amp vary করলে বাকি 2টা constant

FSK (Frequency Shift keying)

BFSK (Binary FSK)

MFSK (Multilevel FSK)

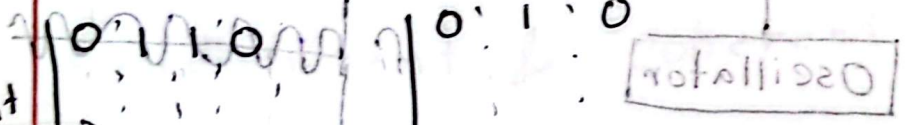
$n_b = 1; 2^1 = 2$

$n_b = 2, 3 \ldots$

Implementation

Non coherent

coherent



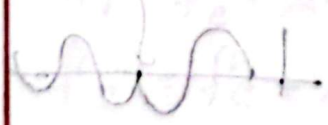
Different phase carry continue হয়ে non coherent

no discontunity
contunity on freq

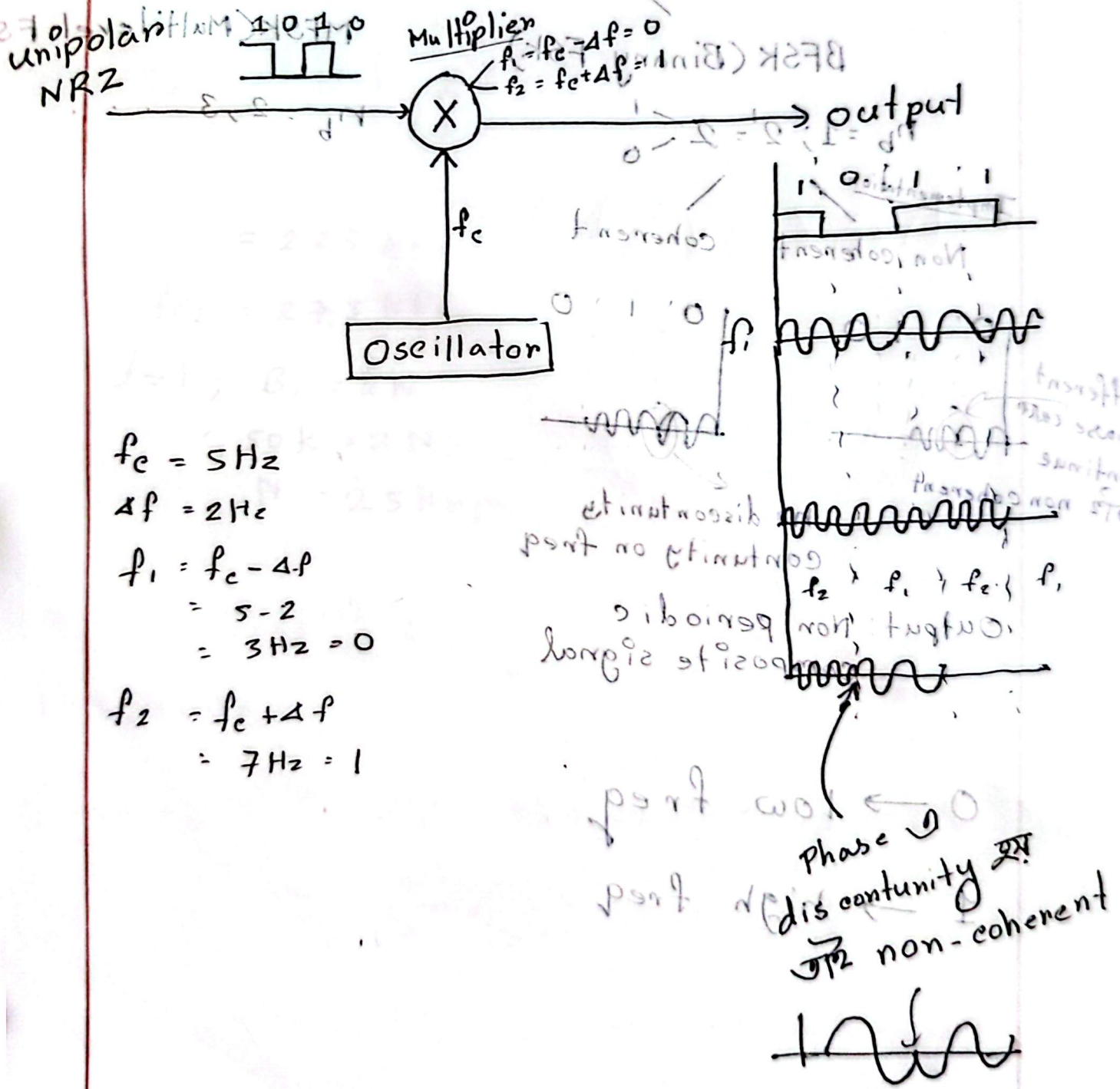
Output: Non periodic composite signal

0 → low freq

1 → high freq



Non coherent Implementation



$f_c = 5\,Hz$

$\Delta f = 2\,Hz$

$f_1 = f_c - \Delta f$

$= 5 - 2$

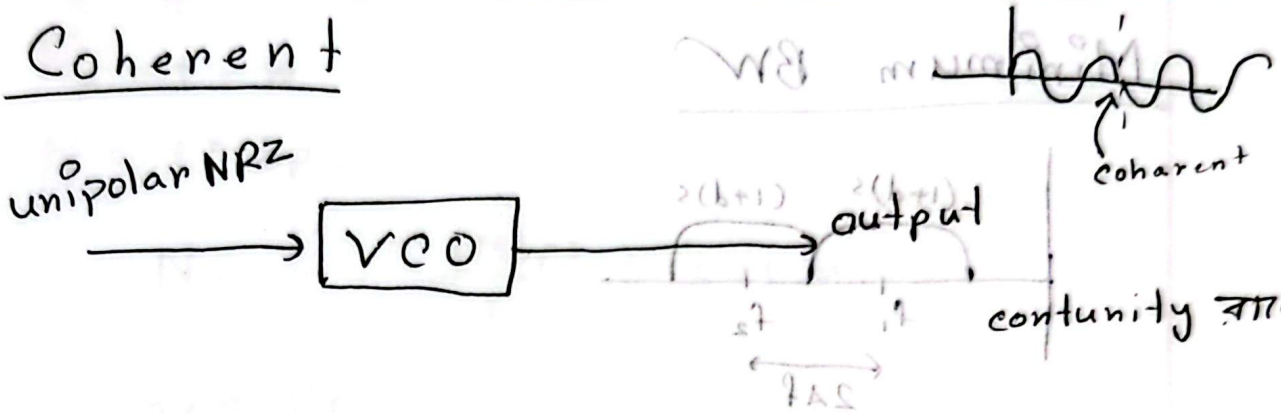
$= 3\,Hz = 0$

$f_2 = f_c + \Delta f$

$= 7\,Hz = 1$

Phase ↓ discontinuity ⇒ non-coherent

Coherent



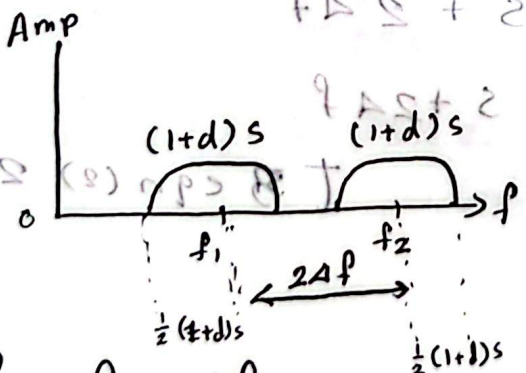
Non Coherent & coherent → non periodic

composite signal

BW analysis of BFSK

Maximum BW

$= B \propto S \; ; \; B = (1+d)S$



$f_1 = f_c - \Delta f$

$f_2 = f_c + \Delta f$

$f_2 - f_1 = f_c + \Delta f - f_c + \Delta f$

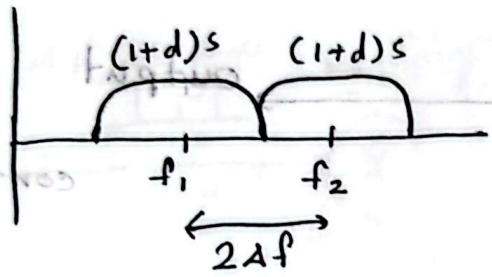
$= 2\Delta f$

$B_{max} = \frac{1}{2}(1+d)s + \frac{1}{2}(1+d)s + 2\Delta f$

$= (1+d)s + 2\Delta f$

Proof

Minimum BW



$$2\Delta f = \frac{1}{2}(1+d)s + \frac{1}{2}(1+d)s$$

$$= (1+d)s$$

$$d = 0 \quad 2\Delta f = (1+0)s = s \quad \text{---(2)}$$

from eqn-1,

$$B_{min} = (1+d)s + 2\Delta f$$

$$= (1+0)s + 2\Delta f$$

$$= s + s$$

$$= 2s$$

Ex - 1

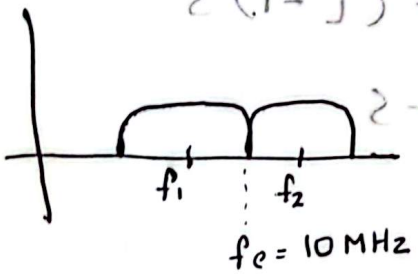
$n_b = 1 = p$

$N = 1\,Mbps$

$S = N \times \frac{1}{r} = \frac{1\,Mbps}{1} = 1\,M\ band$

Minimum BW

$2\Delta f = S = 1\,MHz$



$f_c - f_1 = 2\Delta f = 1\,MHz$

$f_1 = 9.5\,MHz$

$f = 10.5\,MHz$

MFSK

$n_b = 2, 3 \ldots$

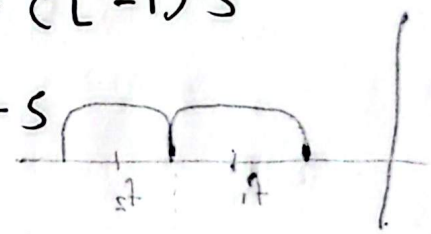
$L = 2^{n_b}$

$B_{max} = (1+d)\,S + (L-1)2\Delta f$

$B_{min} = (1+0)\,S + (L-1)\,S$

$= S + LS - S$

$= LS$



Ex -2

$n = 3$; $L = 2^n = 8$

$N = 3\text{Mbps}$

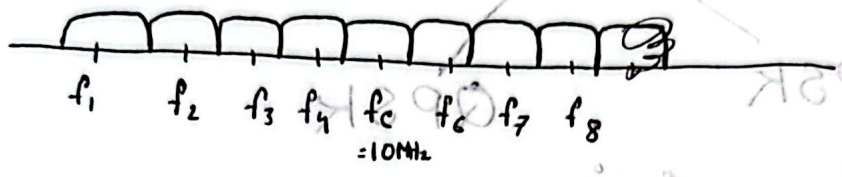
$S = \frac{N}{n} = \frac{3\text{ Mbps}}{3} = 1\text{Mband}$

$2\Delta f = S = 1\text{MHz}$

$f_c = 10\text{MHz}$

Ch-2, 3-2, & 3-3, 4-1, 4-2 ~

analog to digital



$f_5 - f_4 = 2\Delta f = S = 1\ MHz$

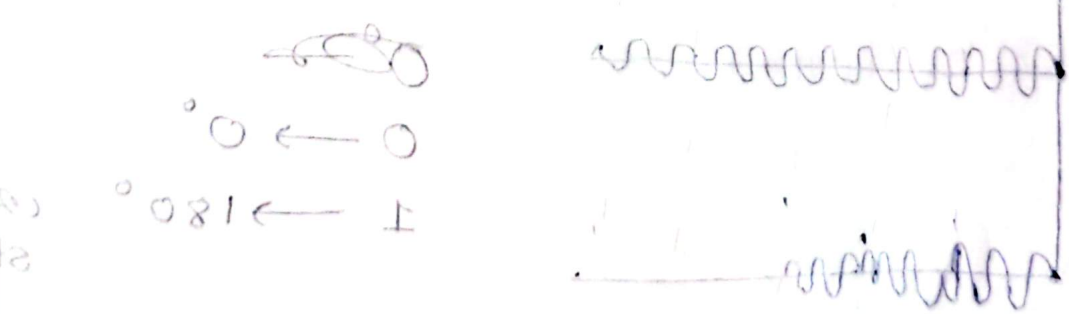
$f_4 = 9.5\ MHz$, $f_3 = 8.5 MHz$

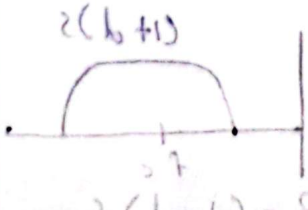
$f_c = 10.5\ MHz$, $f_6 = 11.5\ MHz$

$n_b = 2$; $L = 4$

0	0	$-f_1$
1	0	$-f_2$
0	1	$-f_3$
1	1	$-f_4$

} 4 Level & frequency





Output

Non periodic composite signal

(Phase vary)

PSK

BPSK — QPSK

$n_b = 1$

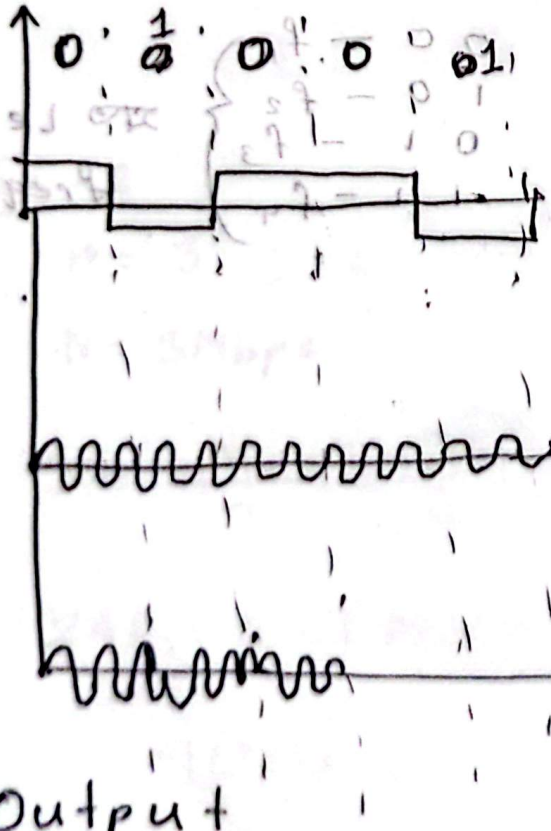
$2^1 = 2$ < 0 - 180°, 1 - 0°

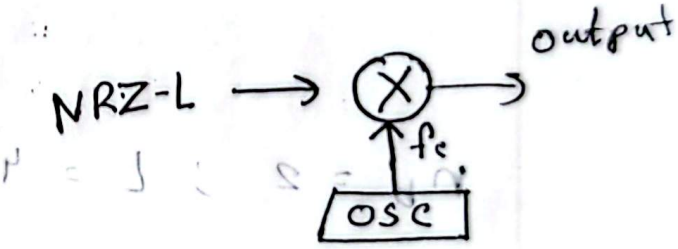
$n_b = 2, 3 \ldots$

$2^2 = 4$

Input

Polar NRZ → NRZ → L





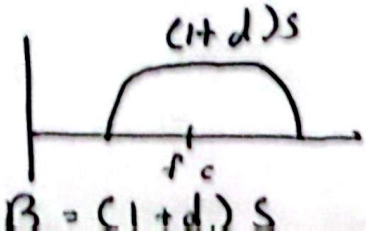
0 → 0°

1 → 180°

start

Output

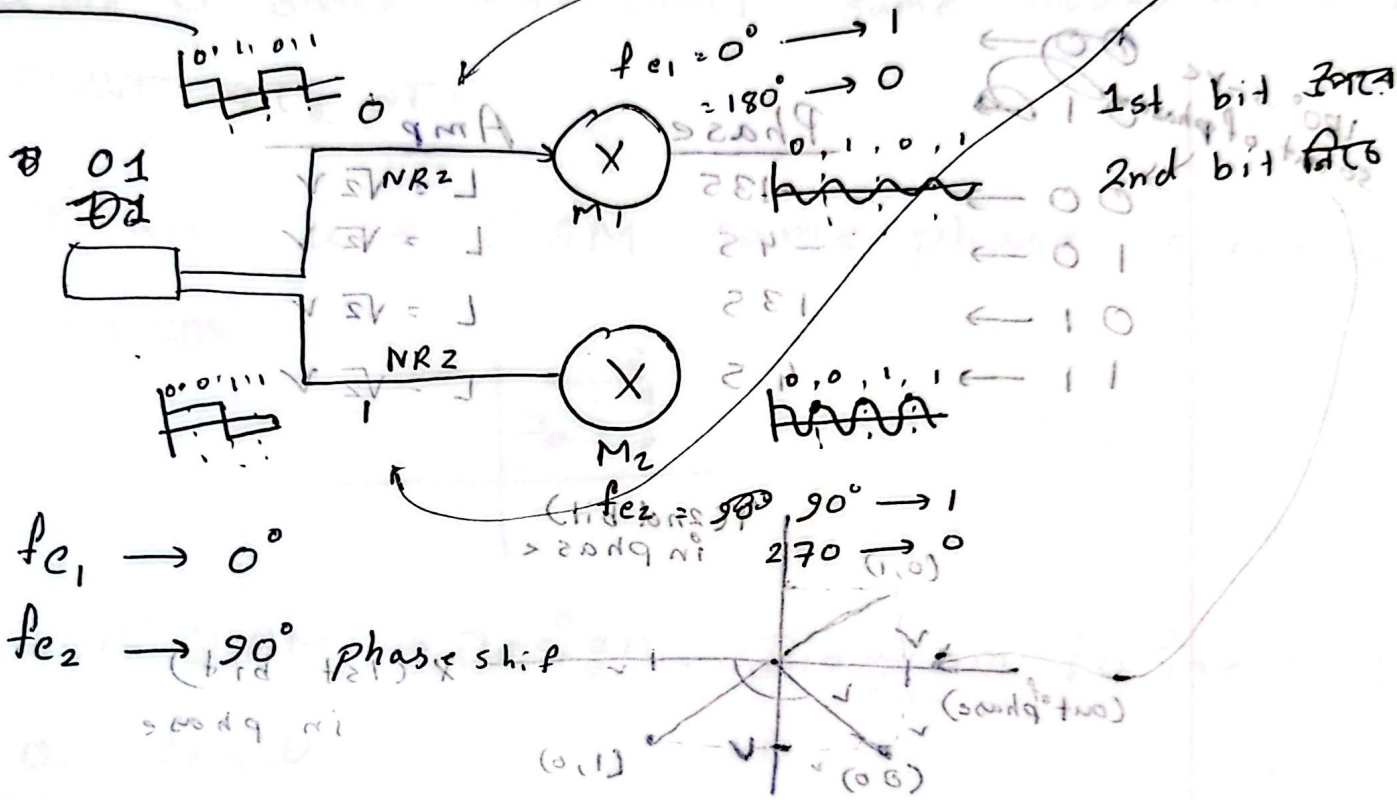
Non periodic composite signal



$n_b = 2$, $L = 4$;

1st: 0 1 0 1
2nd: 0 0 1 1

QPSK



$f_{c_1} \rightarrow 0°$

$f_{c_2} \rightarrow 90°$ phase shift

1st bit

0 - 180°
1 → 0°

2nd bit

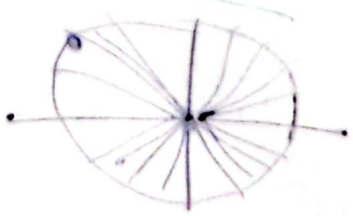
0 - 270°
1 — 90°

00 → ?
10 → ?

this problem is solved by constellation diagram

Phase Amplitude

00 →
10 →
01
11

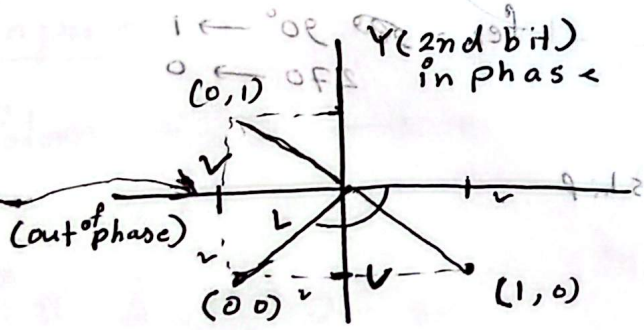


ASK → Unipolar এর 0 1

Phase 180° এর মধ্যে থাকবে

180° → -V
so out of phase

	Phase	Amp
00 →	-135	$L = \sqrt{2}\,V$
10 →	-45	$L = \sqrt{2}\,V$
01 →	135	$L = \sqrt{2}\,V$
11 →	45	$L = \sqrt{2}\,V$



Out of phase

$L^2 = V^2 + V^2$

$= \sqrt{2}\,V$

L - QPSK

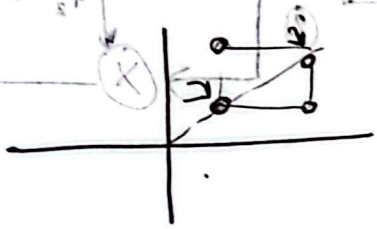
4 - QPSK

16 - QPSK → এক এক টা coordinate এ 4 টা করে।

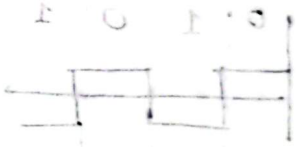
ASK + PSK = QAM

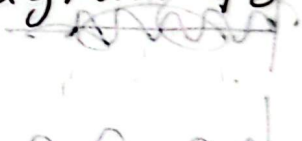
QPSK ~~(QAM)~~ ~~same~~ অনেক বেশী same coordinate ~~এ~~ রাখা যাচ্ছে না।

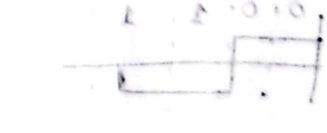
তাই we use QAM. same phase এ multiple bit রাখা যায়



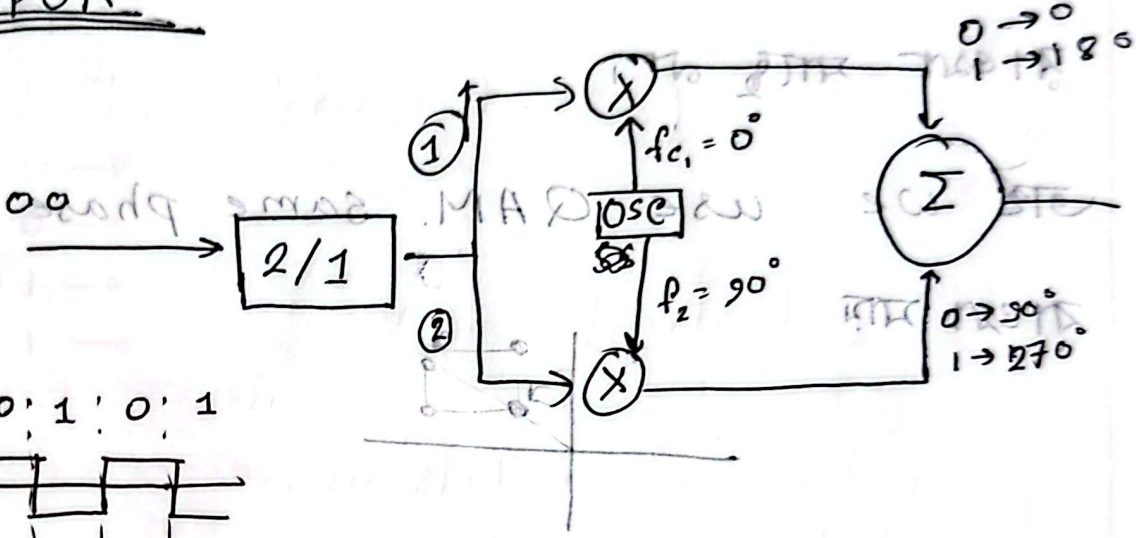
~~QAM~~ QAM → Design. Diagram to identify or draw.

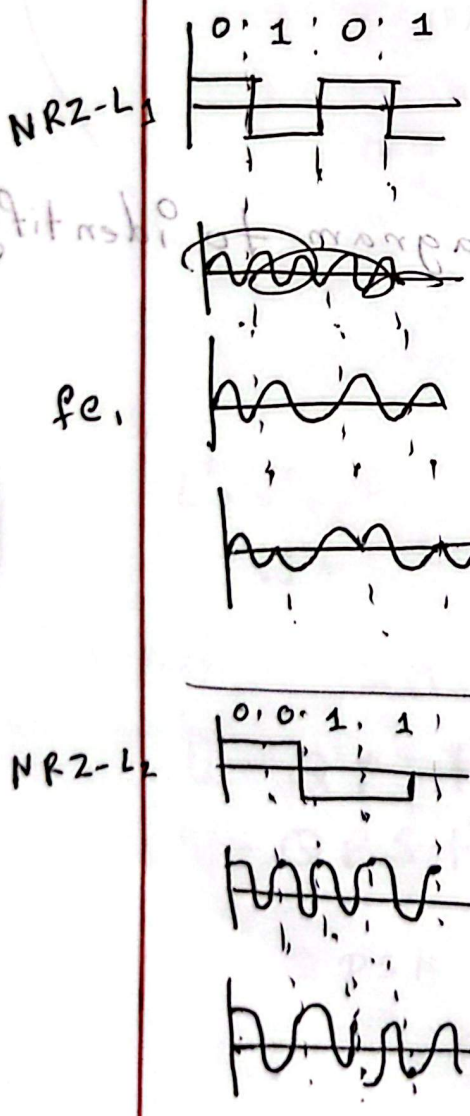






QPSK-


0 → 0°
1 → 180°
2/1
1
2
X
X
OSC
$f_{c_1} = 0°$
$f_2 = 90°$
Σ
0 → 90°
1 → 270°


0 1 0 1
NRZ-L1
fc1
0 0 1 1
NRZ-L2

Quiz-2 (25/8/26)

QAB: same phase & 2 amplitude

4QAB = ~~PKS~~ PSK

5.2

Analog to Analog conversion

Passband channel ও Analog to Analog conversion.

Application on AM. ও in radio.

$$f_c = s(t) = A \sin(2\pi f t + \varphi)$$

- A → Amp (AM)
- f → Freq (FM)
- φ → Phase (PM)

$B_{AB} = 2B$

↑ channel এর bandwidth হবে voice এর bandwidth এর 2 গুন।

f_c always middle এ থাকবে।

AM carrier Freq : 530 ~ 1700 kHz

Q. Total station in AM?

Ans: According to FCC,

AM spectrum range = 530 kHz ~ 1700

Total bandwidth for AM = 1700 - 530
= 1170 kHz

$AM_{BW_{ch}} = 2 \times B_s$

According to FCC,

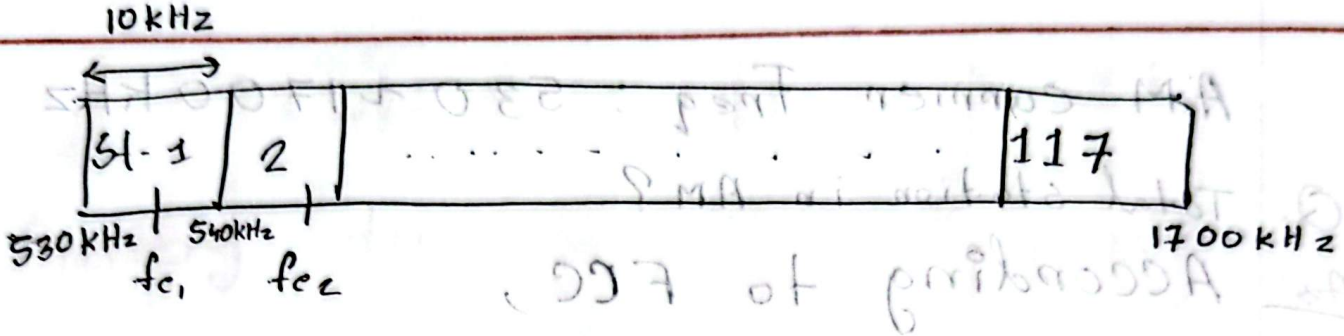
Audio signal bandwidth, $B_s = 5$ kHz

$\therefore AM_{BW_{ch}} = 2 \times 5$ kHz

Number of AM station = $\frac{\text{total BW}}{\text{signal BW}}$

$= \frac{1170 \text{ kHz}}{10 \text{ kHz}}$

$= 117$



So if anyone wants station 1, the spectram will be ~~530 ~ 540 kHz~~

= 530 to 530+10

= 530 to 540 kHz

= 530 ~ 540 kHz

Carrier freq = 530 + 5 kHz

= 535 kHz

General formula;

Station_n = (n-1) × 10kHz + 530kHz

Example,

Station.3 = (3-1) × 10 kHz + 530 kHz

= 20 + 530

= 550 kHz

Spectram range = 550kHz ~ (550+10) kHz

= 550 ~ 560

f_{c3} = 555 kHz

FM:

Positive → high freq

negetive → low .

$$B_{FM} = 2(1+\beta)B_s \qquad \beta = 4$$

According to FCC,

FM ranges from 88 to 108 MHz

Total BW for FM = 108 - 88 = 20 MHz

According to FCC,

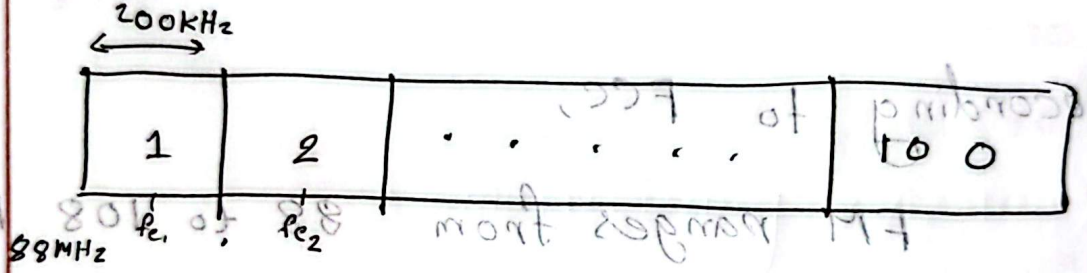
Audio signal BW ; $B_s = 15$ KHz

$$B_{FM_{CH}} = 2(1+\beta)B_s$$
$$= 2(1+4) \times 15$$
$$= 150 \text{ KHz}$$
$$= 150 + \text{Guard band (50KHz)}$$
$$= 200 \text{ KHz}$$

number of FM station = $\frac{\text{total bw}}{\text{Signal Bw}}$

$= \frac{20\ MHz}{200\ kHz} = 100$

100 টি station 50 টি unused



Station-1 spectram range = 88 MHz ~ (88 MHz + 200kHz)

= 88 MHz ~ 88.2 MHz

" carrier freq, f_{c_1} = 88 MHz + 100kHz

= 88.1 MHz

Station-n = (n-1) × 200 kHz + 88 MHz

" -2 = (2-1) × 200 "

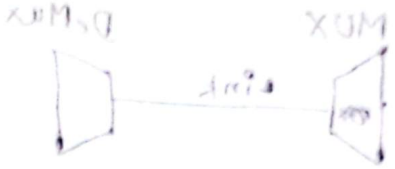
= 200kHz + 88 MHz

= 88.2 MHz

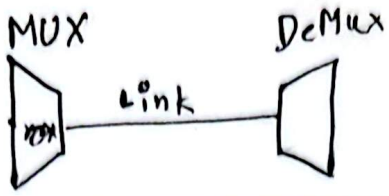
" -2 spectram range = 88.2 MHz ~ (88.2+200)

= 88.2 ~ 88.4 MHz

@ f_{c_2} = 88.2 MHz + 100 kHz = 88.3 MHz



PM: How modulation - working, formula.



Ch-6 (Forouzan)

Multiple signal in a single line.

Mutiple user connected in wifi

↓ line

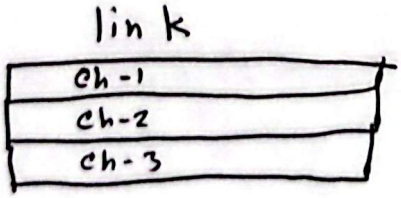
ISP

1) Frequency multiplexing (Analog)

11) Time-division " (Digital)

FDM

1)



channal ची frequency divided

*(Description)

Check for math (MUX, no. of input lines)

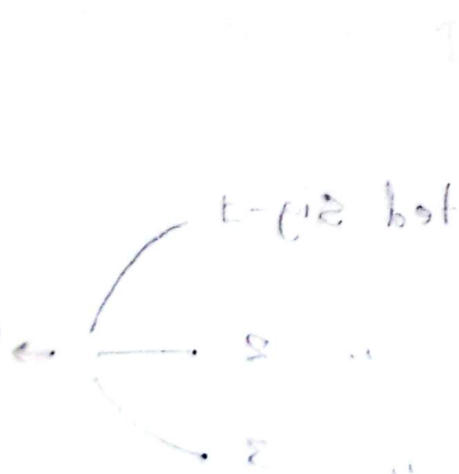
1) Input lines are analog or digital

2) If analog then OK.

3) If digital then convert to analog (ASK, FSK, PSK, QPSK, QAM)

4) Find all input line BW. (BW formula of used convention technique)

5) link BW ≥ input lines BW



Input এর bandwidth and link এর channel এর bandwidth spectrum different তাই Input কে link এর সেই channel দিয়ে pass করতে চাই তার সেই channel এর spectrum এ convert করবো। FM or AM.

Q. Describe multiplexing technique?.

Draw diagram/

$s_1(t) + f_{c_1}$ = Modulated sig-1

$s_2(t) + f_{c_2}$ = " " 2

$s_3(t) + f_{c_3}$ = " " 3

→ Composit signal

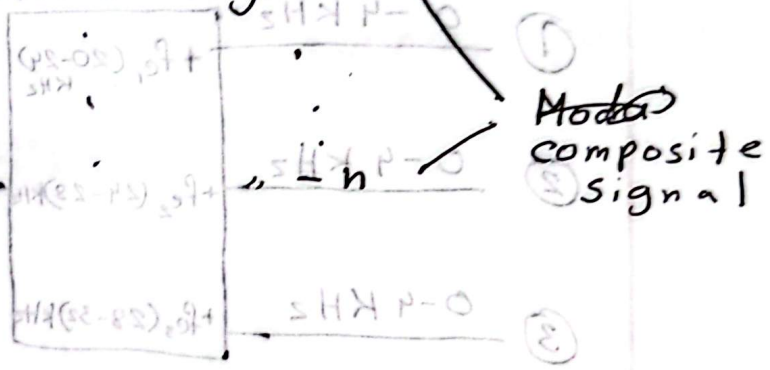
---

have to Draw it.

MUX

$S_1(t) + f_{c_1}$ = Modulated signal-1

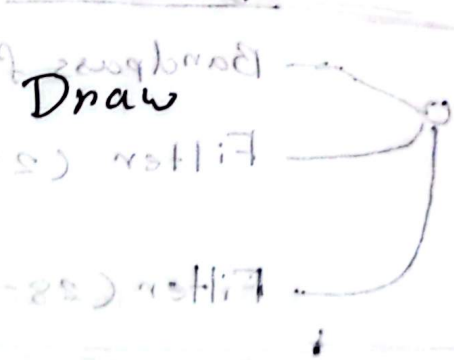
$S_n(t) + f_{c_n}$ = " "



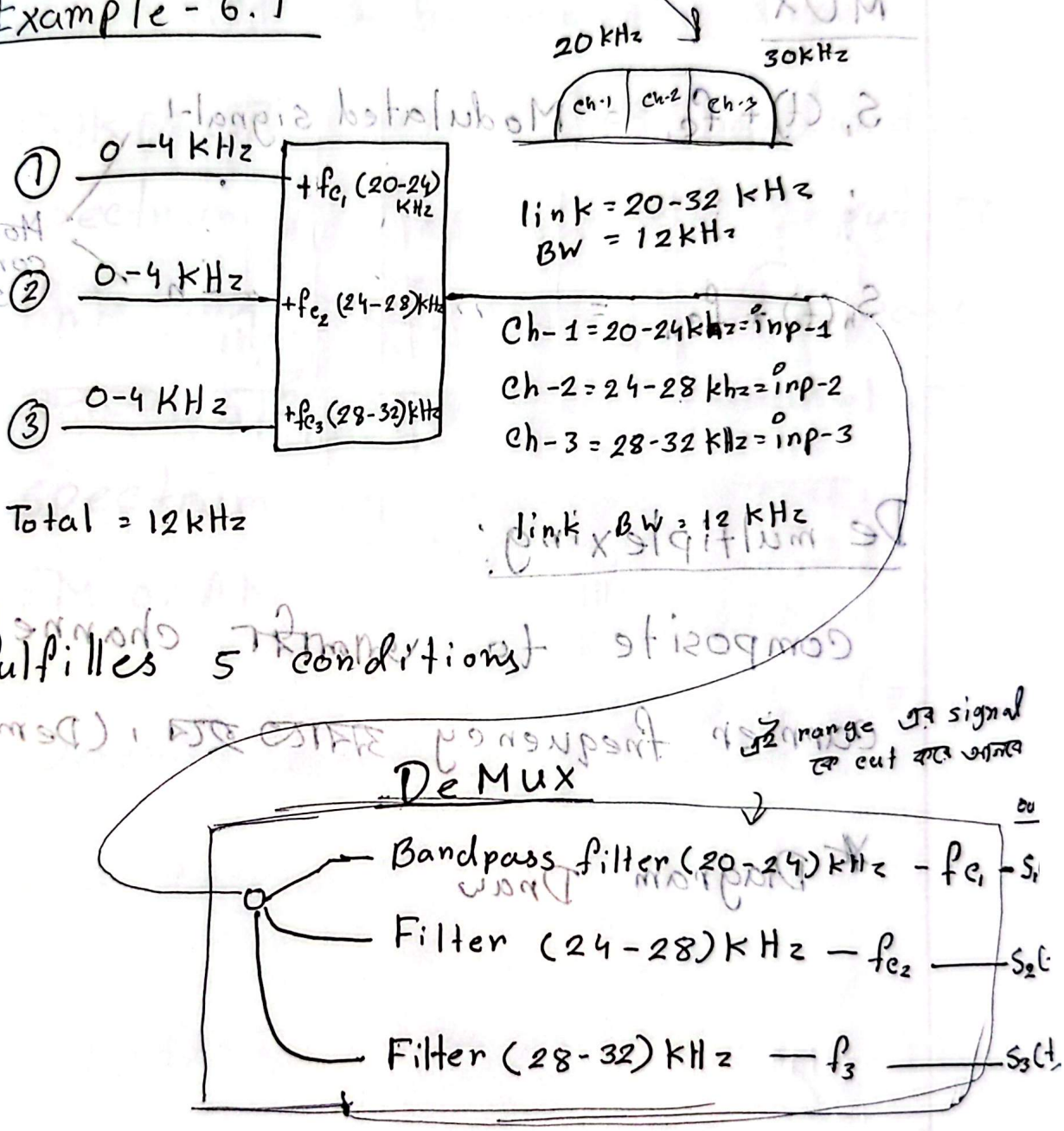
De multiplexing:

composite to আলাদা channel. (Filter carrier frequency আলাদা হবে, (Demodulator)

* Diagram Draw



Example - 6.1



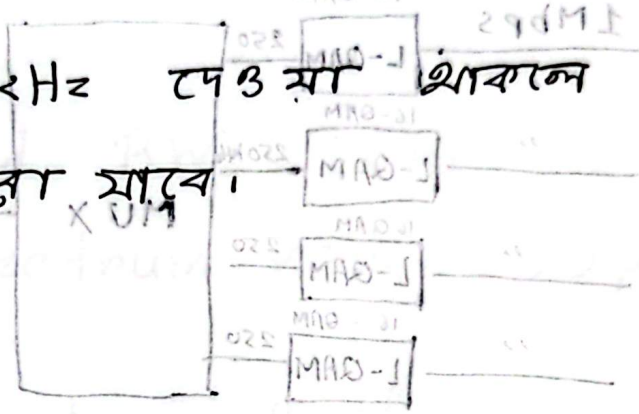
FM use করা হয়েছে।

এখানে link BW = Input BW তাই guard band use করা যায় নাই।

Guard band = 1 kHz 2 টি guard

↓

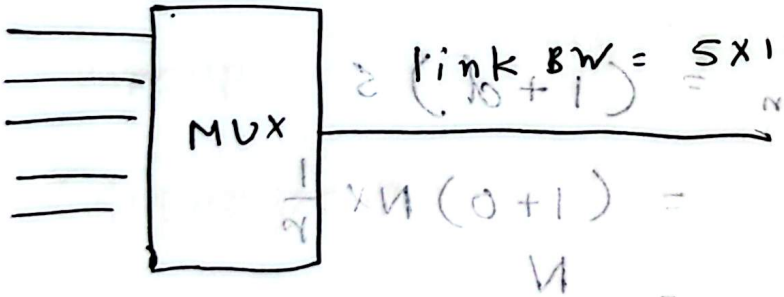
link BW = 12 kHz + 2 × 1 kHz = 14 kHz



link BW = 14 kHz দেওয়া থাকলে guard band use করা যাবে।

Example 6-2

input analogue সিগন্যাল bandwidth দেওয়া।

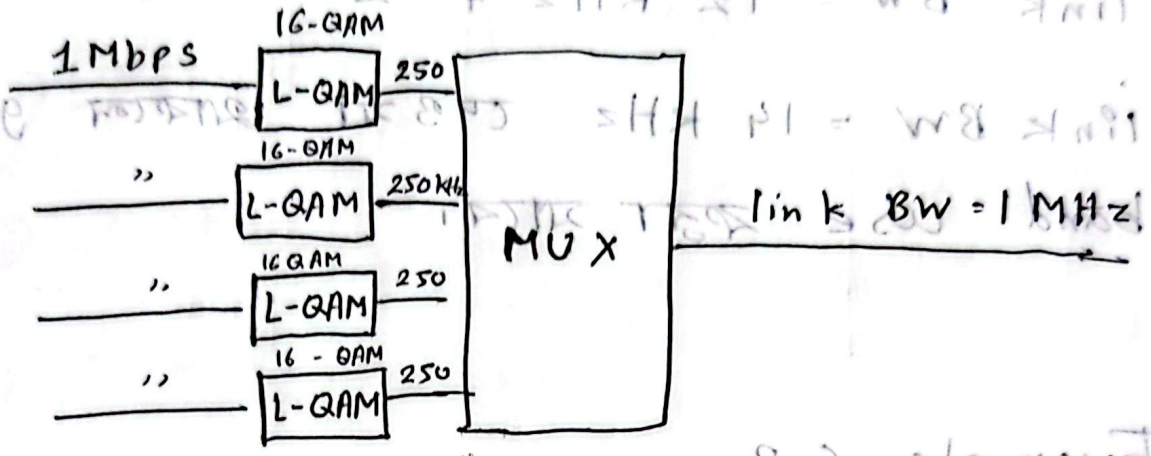


link BW = 5 × 100 + 4 × 10 = 540 kHz

Draw করতে বললে Draw করবো।

Q. এ বলে দিবে QAM use করবে, Level আমাদের find করতে হবে।

6.3



input line BW $= \frac{1\,MHz}{4} = 250\,KHz$

QAM $BW_{min} = (1+d)\,s$

$= (1+0)\,N \times \frac{1}{r}$

$= \frac{N}{r}$

$r = \frac{N}{Bandwidth} = \frac{1\,Mbps}{250\,khz}$

$= 4$

$L = 2^r = 16$

FDM Analog Hierarchy

Q.

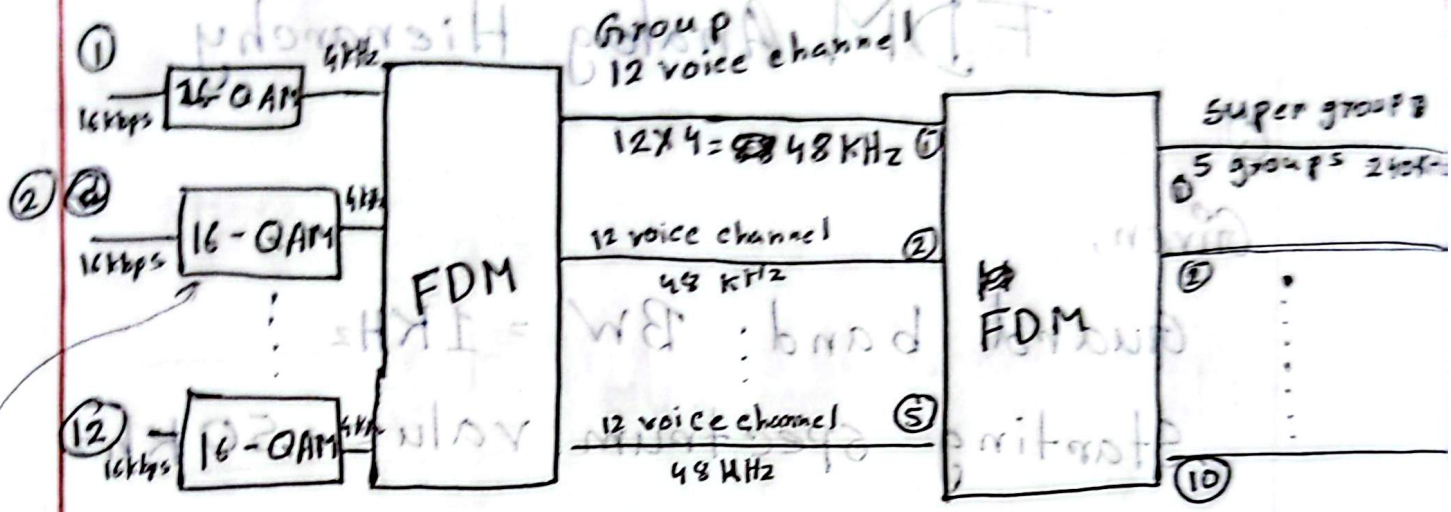
Given,

Guard band BW = 1 KHz

Starting spectrum value = 50 khz

Hiarchy level	# of voice channel
Channel	2 digital signal with data rate 16 kbps
Group	12
Supergroup	60
Master group	600
Jumbo group	3600

by default voice ka BW = 4 kHz



$BW_{min} = (1+d)S$

$= (1+0)S$

$S = N \times \frac{1}{r}$

$r = \frac{N}{B_{min}} = \frac{16\ kbps}{4\ kHz}$

$= 4$ bits / signal elem

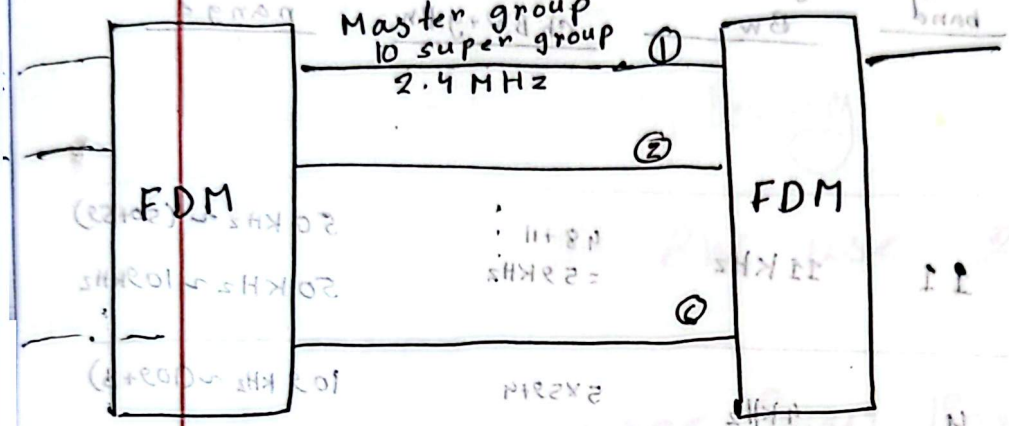
$L = 2^4 = 16$

BW
D to A 4 kHz
12 × 4 = 48 kHz
48 × 5 = 240 kHz
10 × 240 = 2.4 MHz
6 × 2.4 MHz = 13.4 MHz

# of group/supergroup: $\frac{60}{12} = 5$

# of super group/Mastergroup = $\frac{600}{60} = 10$

# of Mastergroup/Jumbogroup = $\frac{3600}{600} = 6$



Hierarchy → Channel | # of voice channel | BW | # of guard band | guardband BW

Hierarchy	# of voice channel	BW	# of guard band	guard band bandwidth	total BW ch BW + g.BW
	11			11 KHz	48+11 KHz = 59 KHz
	4			4 KHz	5 x 59 + 4 KHz = S
	9			9 KHz	10 x S + 9 KHz = M
	5			5 kHz	6 x M + 5 KHz =

Hierarchy level	# of voice channel	BW	# of guard band	guardband BW	Total BW ch BW + g.BW	spectrum range
1	11	11 KHz			48 + 11 = 59 KHz	50 KHz ~ (50+59) 50 KHz ~ 109 kHz
2	4	4 KHz			5 x 59 + 4 =	109 kHz ~ (109+S)
3	9	9 KHz			10x + 9 =	(109+S) ~ (109
4	5	5 KHz			6x + 5 =	

Final total BW = J

Total spectrum Range = 50 ~ (109 + S + M + J)

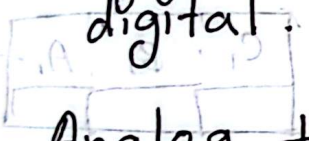
=

TDM

Time ~~फिटर~~ multiplexing.

Total link का BW use हुआ ~~निर्धारित~~ time को.

1. Check to see input lines are analog/ digital.

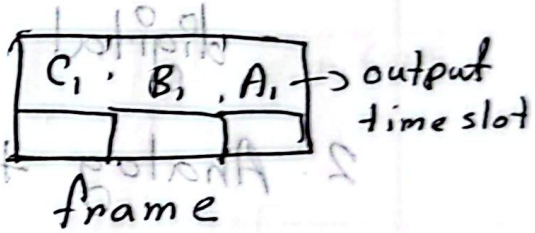


2. Analog then convert to digital signal
   ↳ PCM (sampling Quantization + Encoding)
3. Find data rate of each Input.
4. Check all input data rates are same.
5) If different data rate, then apply data rate management tech.

64 kbps data rate এ কোনো data link এর মাঝে question এ থাকবে, T = হবে,

64 kb যায় 1 sec এ

1 b যায় $\frac{1}{64k}$



1টি frame এ যতগুলো input থাকবে ততগুলো timeslot থাকবে

frame = # of input

frame duration = T

$$\text{output timeslot} = \frac{T}{\#\text{ of input}}$$

~~① input time slot = $\frac{\text{input datarate}}{\text{\# of bits}}$~~

① input time slot = $\frac{1 \times \text{\# of bits}}{\text{input data rate}}$

② output time slot = $\frac{\text{frame duration / input time slot}}{\text{\# of input line}}$

③ frame size = # of output time slot X
# of bits in each output slot

④ output datarate = # of input line X datarate per input line

⑤ frame rate = $\frac{\text{output datarate}}{\text{frame size}}$

⑥ frame duration = $\frac{1}{\text{frame rate}}$ = input time slot

Frame size = 3 X 1

Data rate = 3x input datarate

~~frame size~~

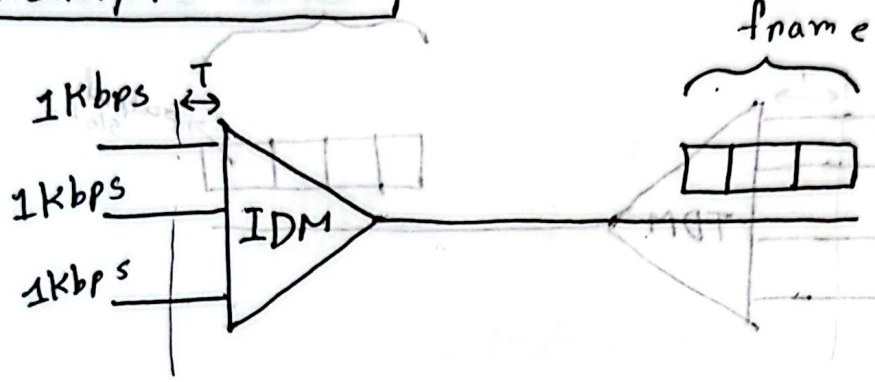
$$\text{frame rate} = \frac{\text{data rate}}{\text{frame size}}$$

→ Synchronous → কোনো input না থাকলেও slot রাখে

→ Statistical → slot রাখে না।

1kbps का digital

Example-6.13



1 bit at a time multiplex की जाएगी।

1 K bits T = 1 sec

1 bits $T = \frac{1}{10^3} = 10^{-3} = 1ms$

Frame duration = input time slot

$= 10^{-3}$

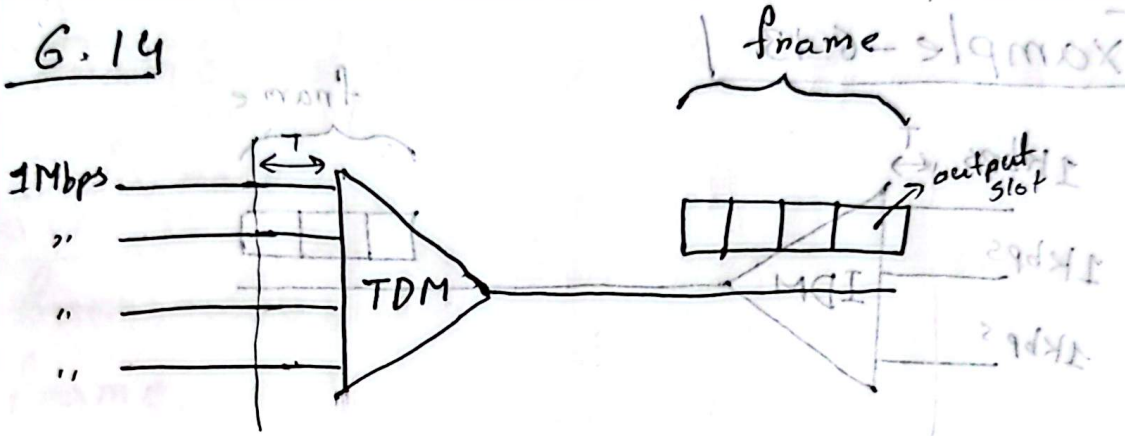
frame size = 3 X 1 = 3 bits

output time slot = $\frac{1}{3}$ ms

output data rate = 3 X 1 kbps = 3kbps

frame rate = $\frac{3kbps}{3 bits}$ = 1 k frames/sec

6.14



Input time slot $= \frac{1}{1\text{ Mbps}} = 1\ \mu s$

frame duration $= T = 1\ \mu s$

output slot $= \frac{T}{4}\ \mu s = \frac{1}{4}$

frame size $= 4 \times 1 \text{ bits} = 4 \text{ bits}$

bitrate/Data rate $= 4 \times 1\text{ Mbps} = 4\text{ Mbps}$

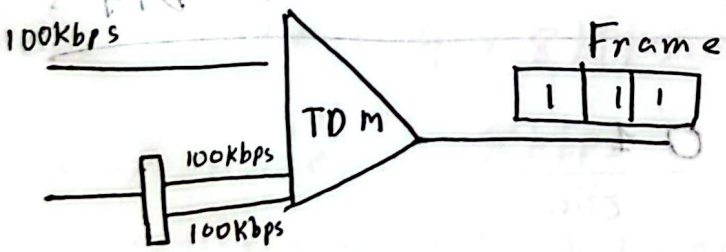
Frame rate $= \frac{4\text{ Mbps}}{4\text{ bits}} = 1\text{ M frames/sec}$

The duration of 1 bit before multiplexing = input time slot

Data Rate Management

- Multilevel → Multiplexing (ADD করা)
- Multishot → De Multiplexing (Divide)
- Pulse Stuffing (fake bit adding) → Adding bits

6.11



$T = \frac{1}{100}$ ms = Frame duration

Frame size = 3 bits

output time slot = $\frac{T}{3} = \frac{1}{300}$ ms

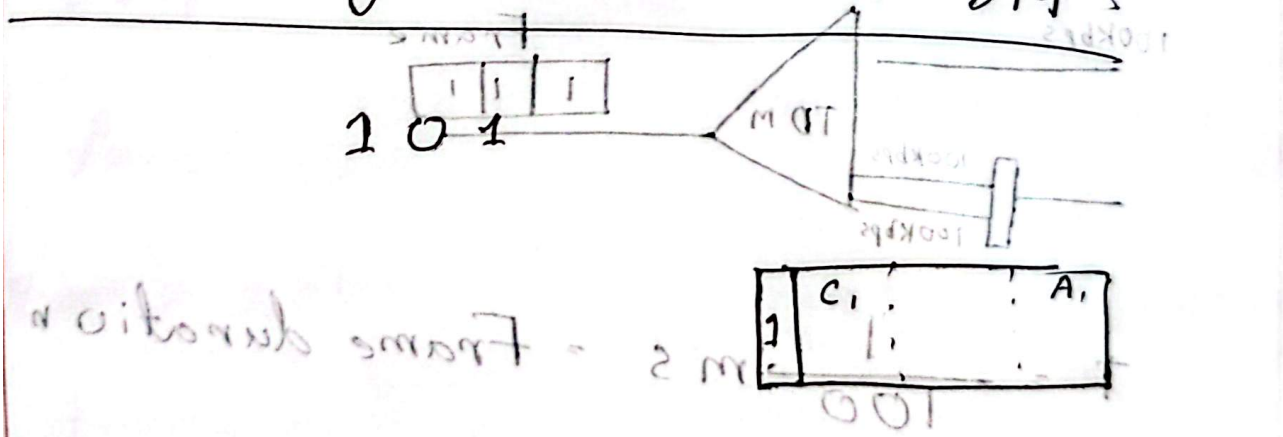
Data rate = 3 x 100 kbps.

= 300 kbps

Frame rate = $\frac{300 \text{ kbps}}{3 \text{ bits}}$

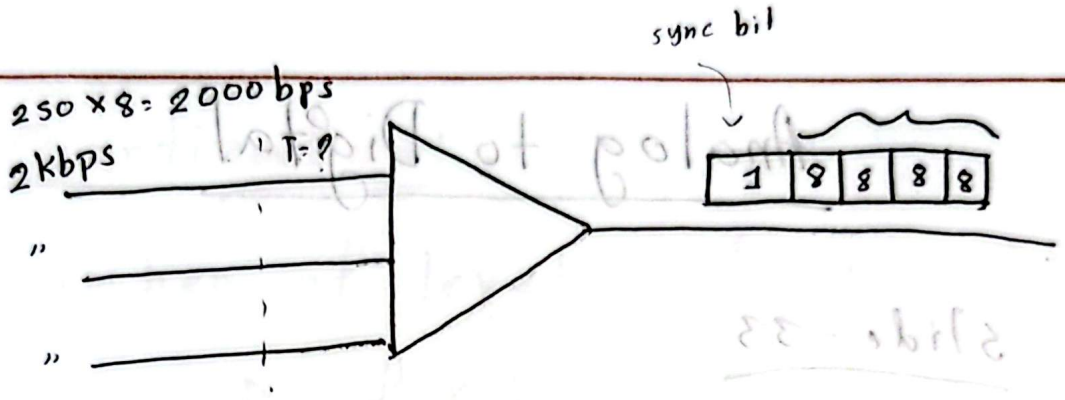
= 100 k frames/sec

Frame synchronization bit :



extra একটা bit add করে পাঠানো হয়।

তখন data rate বাড়ে, so overhead হয়।



$$T = \frac{8}{2\text{kbps}} = 4\text{ms} = \text{input timeslot} = \text{frame duration}$$

data rate = 4 × 2 kbps = 8 kbps

frame size = 4 × 8 bits = 32 bits

number of bit in frame →

$$\text{Frame rate} = \frac{8\text{kbps}}{32\text{bits}} = 250 \text{ frames/s}$$

Frame size with extra bit = 32 + 1 = 33 bits

Data rate = 250 frames/sec × 33 bits

= 8250 bps

Overhead = (8250 − 8000) bps

= 250 bps

Analog to Digital

Slide - 33

$$\frac{1544 \text{ kbps}}{24} = 64.33 \text{ kbps}$$

~~Error~~ fraction means overhead

PCM → Sampling
→ Quantization
→ Encoding

1) Sampling:

According Nyquist theorem,

$f_s = 2 \times f_h$ for voice BW = 4 - 0 = 4

$= 2 \times 4 \text{ kHz}$ $f_h = 4$

$= 8 \text{ kHz}$

$= 8 \text{ k samples/sec}$

8 Kbps overhead এবার example এর মাধ্যমে বের করে.

② Quantization:

number of level ; and bits বের করবো only.

$$N = f_s \times n_b$$

$$\therefore n_b = \frac{N}{f_s} = \frac{64\text{Kbps}}{8\text{K sample/sec}}$$

$$= 8 \text{ bits}$$

$$L = 2^8 = 256 \text{ levels}$$

FDM analogue hyararchy
TDM
} Quiz-3

input line voice → analog
TDM accepts Digital so
PCM needed

Digital Hierarchy

Service	line	Rate (Mbps)	voice ch

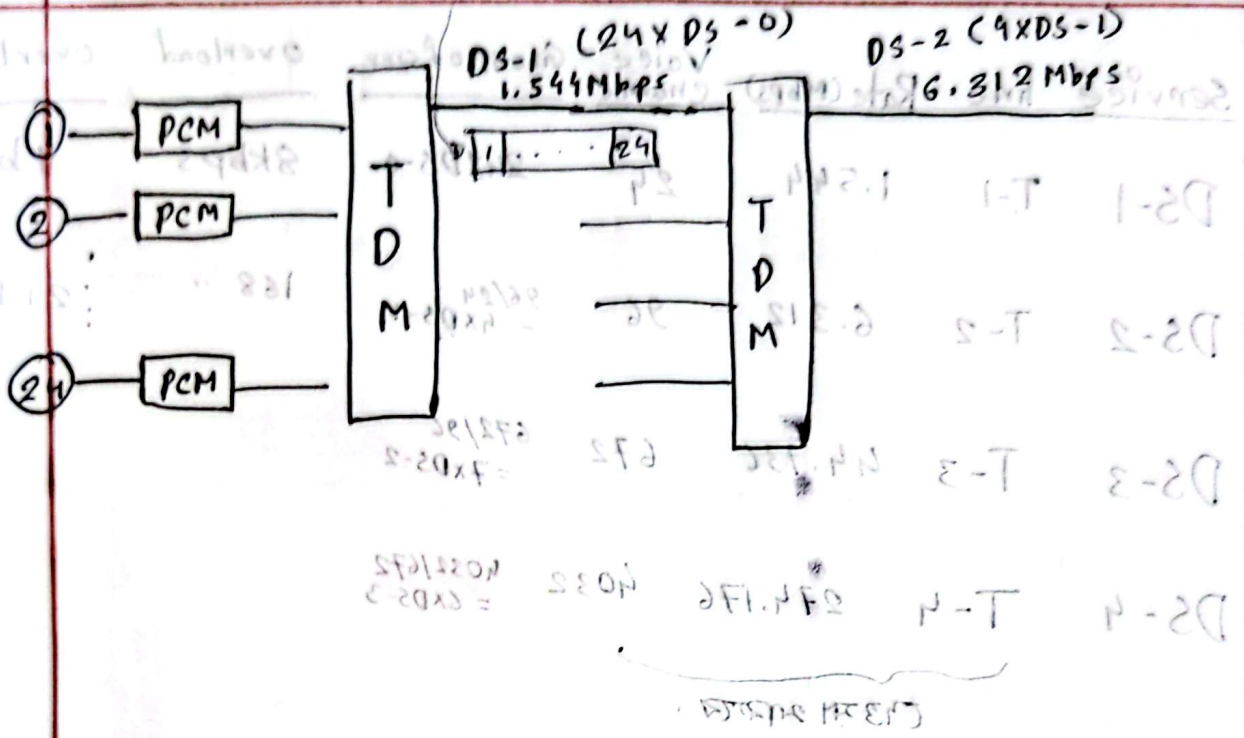
Service line Rate (Mbps) Voice ch Group of service over

TDM analog = PCM
TDM

Service	line	Rate (Mbps)	Voice channle	Group of serv	overload	overload bits
DS-1	T-1	1.544	24	24 x DS-0	8kbps	1 bit
DS-2	T-2	6.312	96	96/24 = 4 x DS-1	168 "	21 bits
DS-3	T-3	44.736	672	672/96 = 7 x DS-2		
DS-4	T-4	274.176	4032	4032/672 = 6 x DS-3		

চেষ্টা করবে।

24 T1 time slot



PCM

① Sampling

$f_s = 2 \times f_h = 2 \times 4\,K$

$= 8K$ samp/sec

② Quantization

$N = f_s \times n_b$

$n_b = \frac{N}{f_s} = \frac{64k}{8k} = 8$ bits

$L = 2^{n_b} = 2^8 = 256$

$\frac{1.544}{24} =$

DS-1 = 24 × 8DS-0
= 24 × 64 kbps
= 1536 "

According to table,

DS-1 data rate = 1544 kbps
overhead = (1544 − 1530) kbps
= 8 kbps

without overhead,

Frame size = 24 × 8 bit
= 192 bit

frame rate = $\frac{1536}{192 \text{ bits}}$
= 8k frames/sec

overhead বাদ

overhead bits = $\frac{\text{overhead}}{\text{Frame rate}}$

$= \frac{8k}{8k} = 1 \text{ bit}$

with overhead,

Frame size = (192 + 1) bits = 193 bits

Data rate = 8k × 193 bits

= 1.544 Mbps

DS-2 = 96 × DS-0 = 4 × DS-1

= 96 × 64 kbps

= 6144 kbps

According to table,

DS-2 data rate = 6312 kbps

overhead = (6312 - 6144) kbps

= 168 kbps

without overhed,

Frame size = 96 × 8 bits

= 768 bits

$$\text{frame rate} = \frac{6144\text{k}}{768 \text{ bits}} = 8\text{k frames/sec}$$

$$\text{overhead bits} = \frac{\text{overhead}}{\text{frame heat}} = \frac{168\text{K}}{8\text{K}} = 21 \text{ bits}$$

with overhead,

Frame size = 768 bits + 21 bits

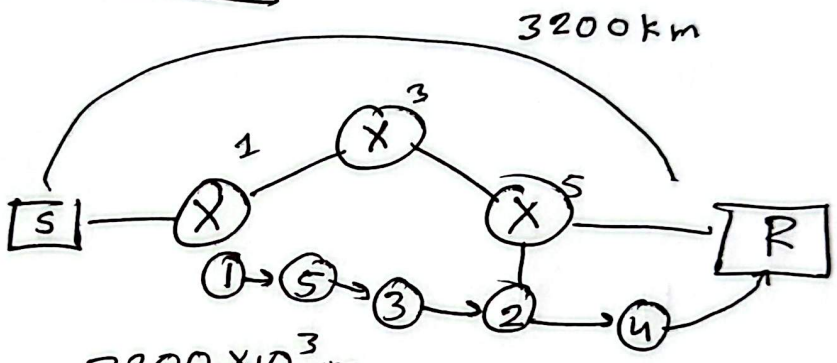
= 789 bits

Data rate = 8k × 789 bits

= 6.312 Mbps

datagram → individual packet

Problem-2



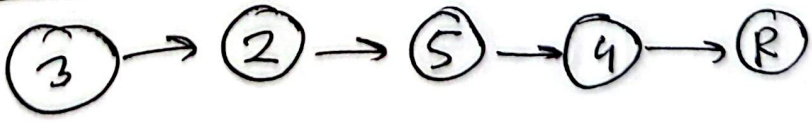
$$P_d = \frac{3200 \times 10^3 \text{ m}}{2 \times 10^8 \text{ m/s}} = ?$$

$$T_d = 0$$

$$\text{delay} = P_d + W_1 + W_3 + W_5$$

$$= \frac{D}{5} + 3 + 20 + 20$$

flow must after examining:



VC

Datagram circuit sw

setup = req + acknowladge

VCI → mac address

Setup phase:

assigned by switch

Port	VCI	Port	VCI
1	14	3	

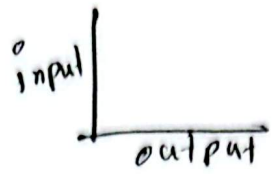
decided by global table

Global table

Dest	port

VCI = 77 assigned by receiver

Teardown phase: local table remove



2 type switch design → আছে

Types of switch

crossbar:

এই আগে output এর আগে কথা বলতে পারে।

non blocking → সবাই সবার আগে communicate করতে পারে।

number of crosspoint = n x m = 3 x 4 = 12
↳ transistor

= 1000 x 1000 = 1000000

objective: ① non blocking system design করতে হবে
② number of crosspoint reduce

solution: Multistage switch ✱ (design আসতে পারে)

$N = 200 \times 200$

$n = 20$

$k = 4$

$\frac{200}{20} = 10$ $\quad k = 4$ $\quad N/n = 10$

20×4 $\qquad$ 4×20

10×10

10×10

20×4 $\qquad$ 4×20

1st stage

# of crosspoint $= 10(20 \times 4) = 10 \times 80 = 800$

2nd stage

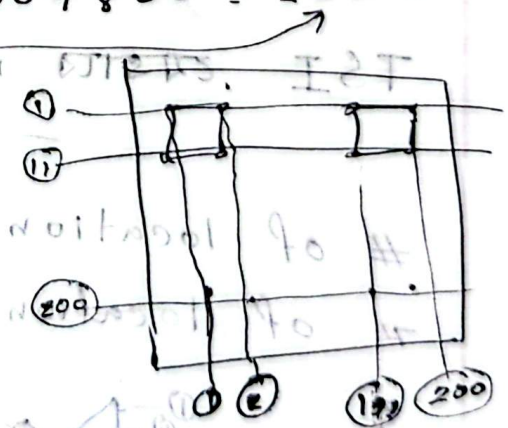
# of crosspoint $= 4(10 \times 10) = 4 \times 100 = 400$

3rd stage

" " $= 10(4 \times 20) = 10 \times 80 = 800$

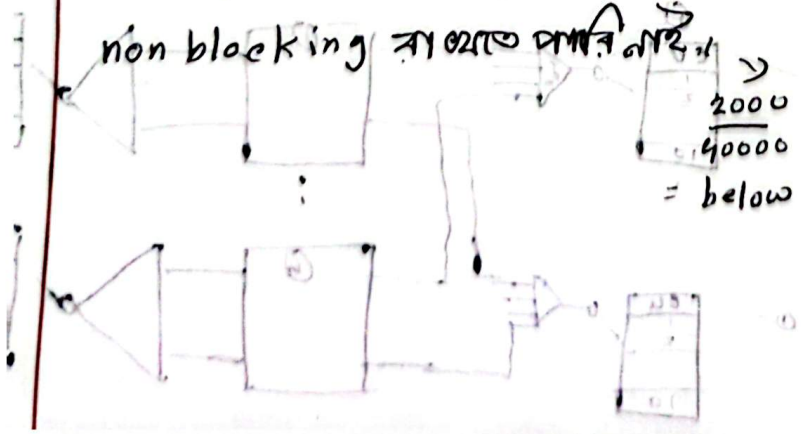
Total crosspoint $= 800 + 400 + 800 = 2000$

if single crossbar used



crosspoint কমানোর

non blocking বানাতে পারি না



2000

40000

= below 25% so blocking

# of crosspoint = 200×20

$= 40000$

forozan এর book

So non blocking এর জন্য

k> =

10 10X19 19X10

20X20

20X20

10 10X19 19X10

1st stage

= 20X190

2nd stage

3rd stage

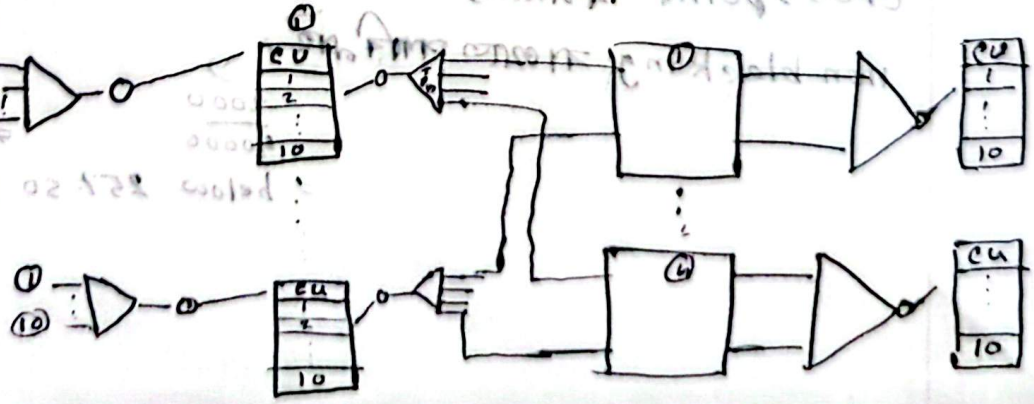
20X190

Total = 11520

TSI যেভাবে mention করবে, সেভাবে যাবে।

# of location = # time slot

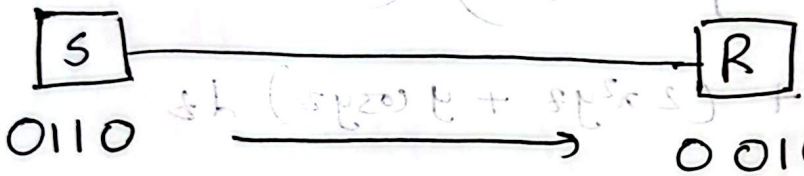
# of location size = # of bits / time slot



Doom

Ch-10

Error detection & correction



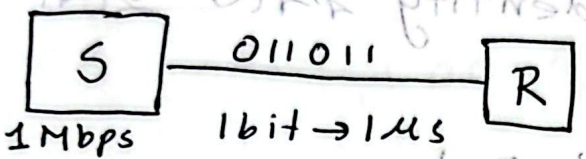
wrong info

during transmission : noise এর কারণে

Error

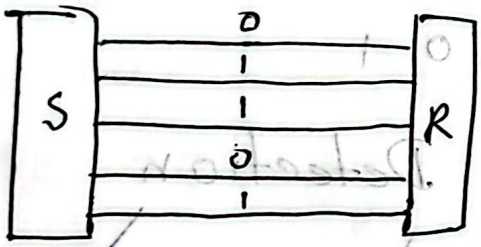
- Single bit error
  - Parallel transmission
- Burst error
  - Serial transmission

Serial transmission



noise যদি 1μs এর বেশী থাকে তখন একাধিক bit effect করে।

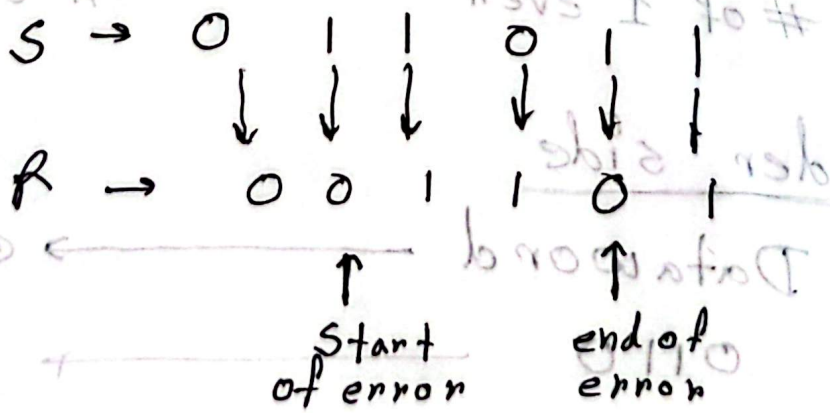
Parallel transmission



1μs এর বেশী noise stay করলে 1 টার বেশী bit কে effect করানো যায় না।

Q. what is error, type of error, why signle bit error in serial transmission?, why burst error in parallel transmission

Burst error :



length = 4

error correction এ extra কোন bit change হয়েছে position identify করতে হবে।

sender → 0 11 0 1

redundency → 0 11 0 1 [1] ← for error detection

reciever → 0 11 0 1

Error Detection

- Parity check
- CRC
- Checksum

Parity check (VRC) (vertical redundency check)

(Mention করতে হবে) Even parity
- # of 1 even

ODD parity
- # of 1 odd

Sender side

Dataword → codeword

0110 → 01100 ← even no of 1

1000 → 10001 ← even no of 1

Even

ODD parity sender side

0110 ⟶ 01101

1000 ⟶ 10000

Reciever side for odd

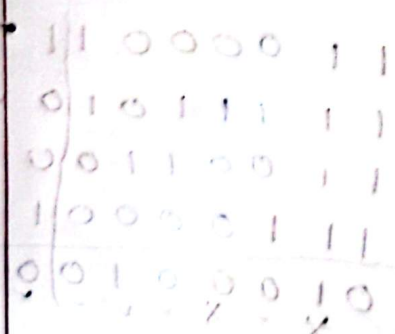
code word ⟶ Error detection

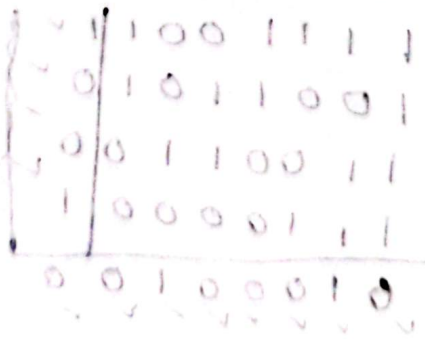
01101 ⟶ ⊕ : no error

100091 ⟶ error কারণ even হয়ে গেছে।

Parity check (VRC):

- even number of error হলে detect করতে পারে না;
- odd number of error হলে detect করতে পারে।
- good for single bit error





2-D parity check (Even)

Data word

ASCII কে binary তে নিচে দেখানো হচ্ছে,

1111001 1001101 1100110 1110000

1111001	1
1001101	0
1100110	0
1110000	1
0100010	0

← row wise even parity

↑ col wise even parity generate

codeword:

1111 0011 1001 1010 1100 1100 1110 0000 1 01000100

receiver: codeword

1111001	1 ✓
1001101	0 ✓
1100110	0 ✓
1110000	1 ✓
0100010	0 ✓

✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓

no error

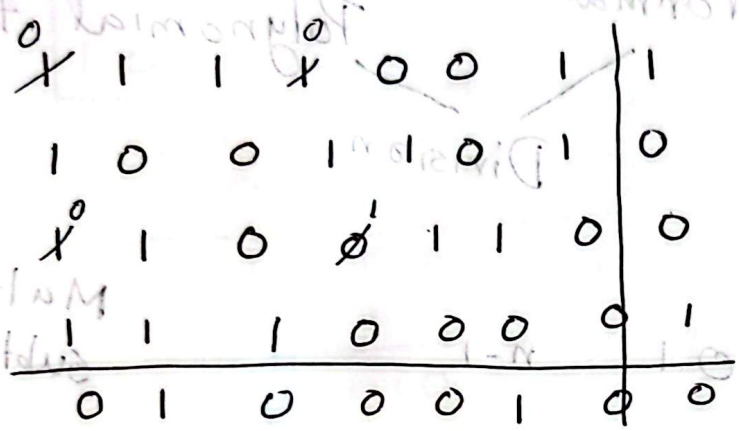
Modified if error

1 1 0000	1 ✓
1 1 1101	0 ✓
1 1 0011	0 ✓
1 1 0000	1 ✓
0 1 0001	0 ✓

✓ ✗ ✓ ✗ ✓ ✓ ✓ ✓

2-D parity check

limitation



2টা row এর same position এ error হলে identify করতে পারে না।

CRC (cyclic redundency check)

Binary Format — Polynomial format

Division

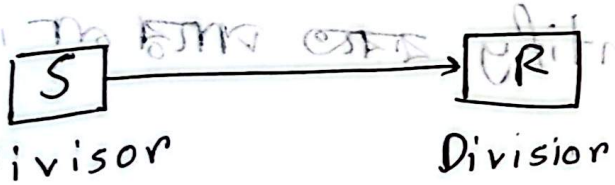
Binary:

Data = 1001 — n-1

Divisior = 1011 — n

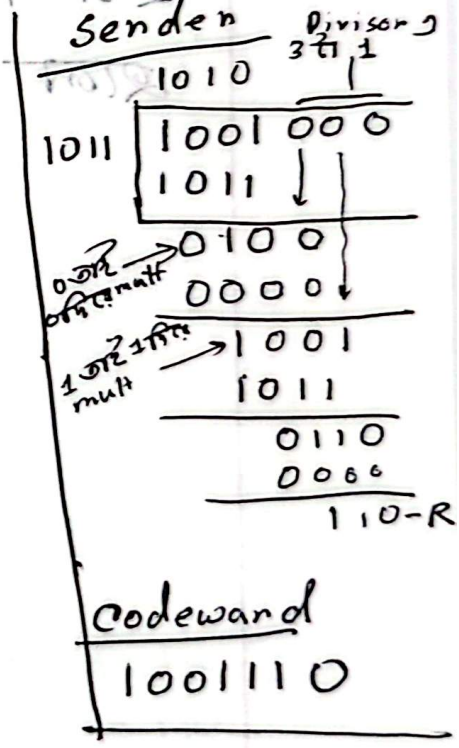
Mult → And op

Subt → XOR op

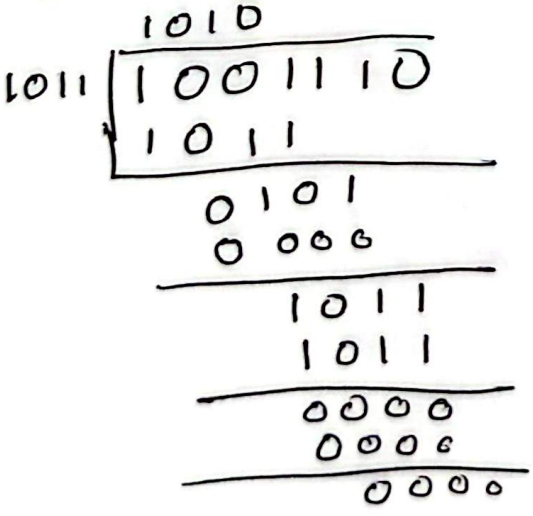


Data → codeward

Sender



reciever



no error

if recieved with error

```
1011 | 1 1 0 1 1 0
     | 1 0 1 1
     -----------
       1 1 0 1
       1 0 1 1
       -------
         1 1 0 1
         1 0 1 1
         -------
           1 1 0 0
           1 0 1 1
           -------
             1 1 1   x error
```

most efficient

→ Burst + single error both handled

Polynomial

Data = 1001

$1 \times x^3 + 0 \times x^2 + 0 \times x^1 + 1 \times x^0$

Data = $x^3 + 1$

Divisor = 1011

$1 \times x^3 + 0 \times x^2 + 1 \times x^1 + 1 \times x^0$

$= x^3 + x + 1$

Divisor ka highest term multiply

Data = $(x^3 + 1)\, x^3 = x^6 + x^3$

$x+1 \;\overline{)\; x^6 + \quad + \quad + x^3 \;}$

$x^6 + \quad + x^4 + x^3$

x^4

$x^4 + x^2 + x$

$x^2 + x$

code word = $x^6 + x^3 + x^2 + x$

~~Sender~~
reciever

x^3+x+1) $x^6 + x^3 + x^2 + x$ ($x^3 + x$

$x^6 + x^4 + x^3$

$x^4 + x^2 + x$

$x^4 + x^2 + x$

0 no error

Check Sum

step-1 বলে দিবে →

Sender side – 4bit (0-15) step-2

Data - 7, 11, 12, 0, 6 codeword = 7, 11, 12, 0, 6, 9

7
11
12
0
6
0

always 0

Sum = 36

32 16 8 4 2 1

36 = 1 0 0 1 0 0

extra → 1 0

wrapped sum = 0 1 1 0 = 6

check sum = 1 0 0 1 = 9

reciever

codeword = 7, 11, 12, 0, 6, 9

7
11
12
0
6
9

sum = 45

32 16 8 4 2 1

45 = 1 0 1 1 0 1

1 0

wrapped sum = 1 1 1 1

check sum = 0 0 0 0 → no error

received with error

codeword = 7, 11, 12, 1, 6, 9

```
   7
  11
  12
   1
   6
   9
 ----
```

sum = 46

```
46 =  32  16   8   4   2   1
      1   0   1   1   1   0
      ~~~~~~~~~~~~~~~~~~~~~~
                      1   0
      ----------------------
          1   0   0   0   0
      ~~~~~~~~~~~~~~~~~~~~~~
      ----------------------
          0   0   0   1
```

wrapped sum = 0 0 0 1

check sum = 1 1 1 0 → error

add এমনে last 0 zero

এটা দিলে ASCII count

16 bit (F=70, o=111, r=114, u=117, z=122, a=97, n=110)

Forouzan

Fo = 4 6 6F

ro = 7 2 6F

uz = 7 5 7A

an = 6 1 6E

0 0 0 0

F = 70/16 = 16)70 (4 = $(46)_{16}$
64
6

o = 111/16 = 16)111 (6 = $(6F)_{16}$
96
15(F)

r = 114/16 = 16)114 (7 = $(72)_{16}$
112
+5(F)

1 1 2

4 6 6F
7 2 6F
7 5 7A
6 1 6E
0 0 0 0
8 F C 6
1
wrapped sum = 8 F C 7

sum = wrap

check sum = 70 3 8

15 দিয়ে বিয়োগ

15
15
10
14
0
sum: 54 = 16)54 (3
48
6
= 6

0 1 3
4 6 6
7 2 6
7 5 7
6 1 6
0 0 0
24 15(F) 28

= 16)24 (1
16
8

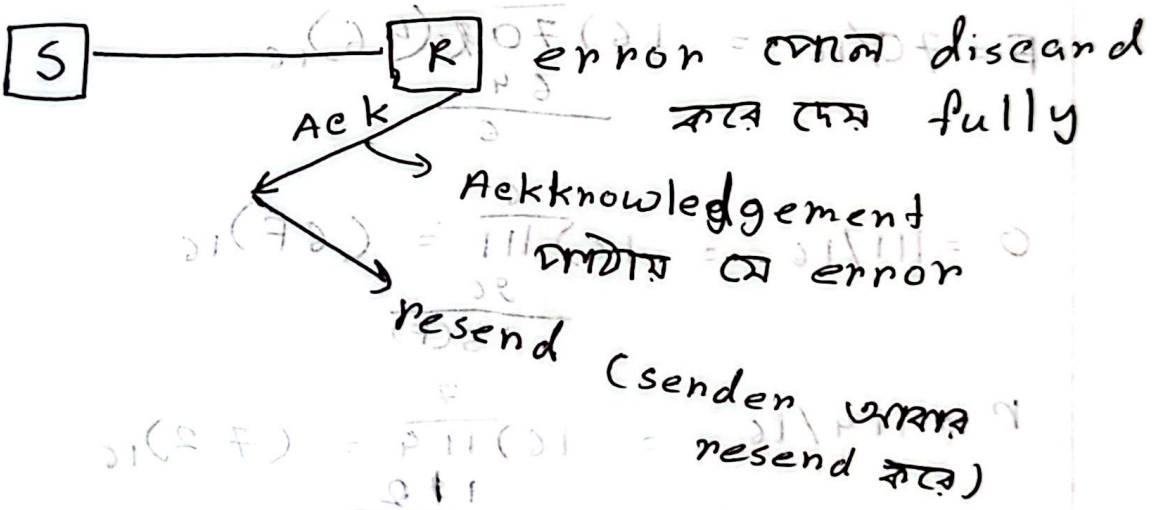
= 16)28 (1
16
12

7 7 7 8 8 0
27

codeword = Forouzan, 0_x (7038)

receiver এ check sum zero আসলে no error

Error correction



Reciever direct error correct করলে সেটা FEC (using hamming code)

Error correction (Hamming code)

Sender side

Data ⟶ codeword

Data = 1001101

k = 7 bits

redundant = r = ? = 4 bits

codeword = k + r

= 7 + 4

= 11 bits

$2^r > k + r + 1$

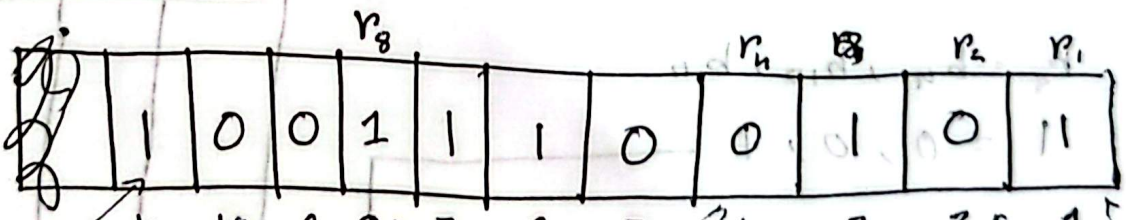
r = 4 দিলে,

$16 = 2^4 > 7 + 4 + 1 = 12$

∴ r = 4

Codeword

r বসাবে 2 এর power এর জায়গায়



r বারে বারি যায়গায় Data

1টা বাদ দিয়ে দিয়ে

2টা নিবে 2টা বাদ

$r_1 = b_3, b_5, b_7, b_9, b_{11}$

$= 1, 0, 1, 0, 1$ = even no. of 1 এর জন্য 1 add

$r_2 = b_3, b_6, b_7, b_{10}, b_{11}$

$= 1, 1, 1, 0, 1$ = even এর জন্য 0 add

$r_4 = b_5, b_6, b_7$ | $r_8 = b_9, b_{10}, b_{11}$

$= 0, 1, 1$ | $= 0, 0, 1 = 1$

codeword:

1 0 0 1 1 1 0 0 1 0 1

sent with error

1 0 0 1 1 1 0 0 1 1 1

$r_1 = b_3, b_5, b_7, b_9, b_{11}$

$1 = 1, 0, 1, 0, 1$

$r_2 = b_3, b_6, b_7, b_{10}, b_{11}$

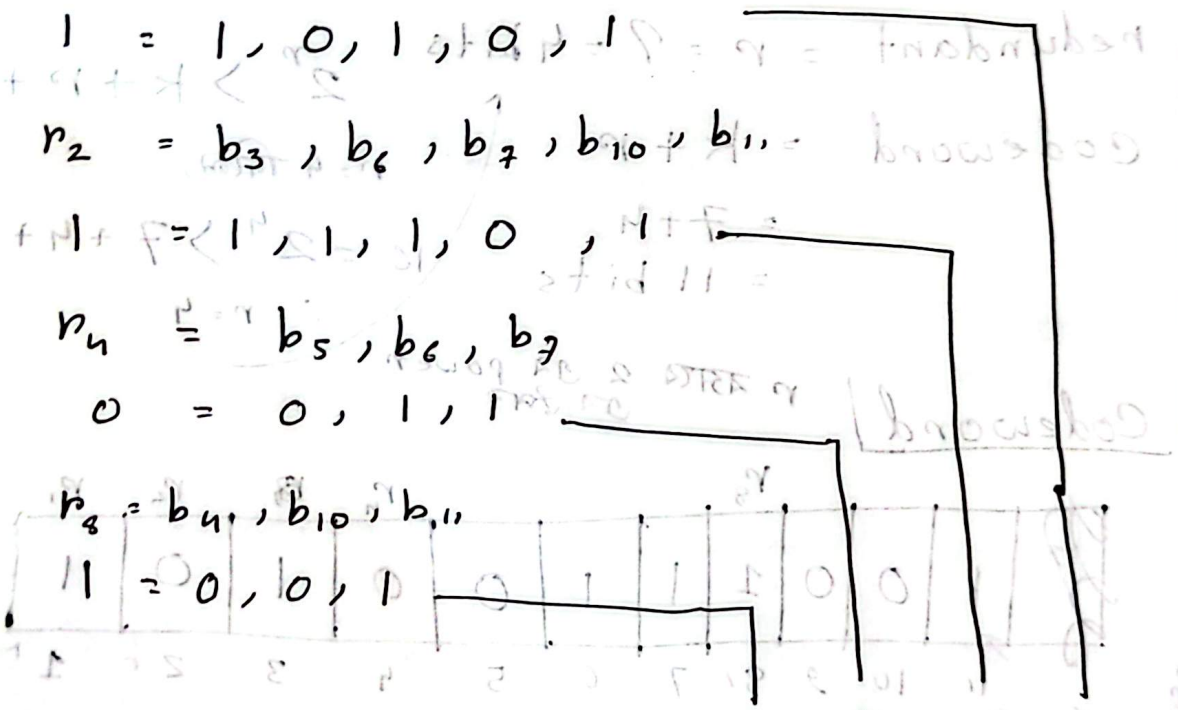
$1 = 1, 1, 1, 0, 1$

$r_4 = b_5, b_6, b_7$

$0 = 0, 1, 1$

$r_8 = b_9, b_{10}, b_{11}$

$1 = 0, 0, 1$



position of error পেয়ে গেছি,

corrected code : 2 no. pos কে invert. r এর position বাদ দিলে Data থাকবে

1 0 0 1 1 1 0 0 1 0 1